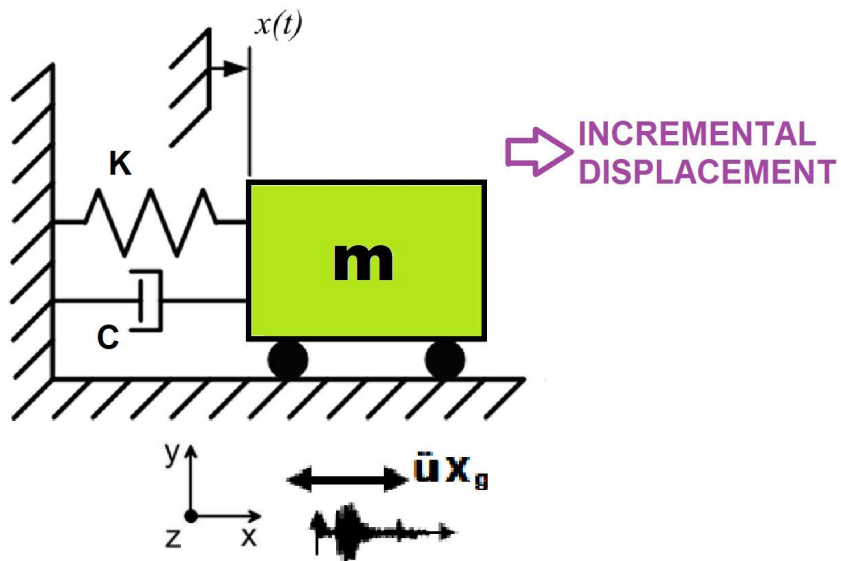


>> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<

# COMPREHENSIVE NONLINEAR SEISMIC ASSESSMENT OF A SINGLE-DEGREE- FREEDOM (SDOF) STRUCTURE: AN OPENSEES FRAMEWORK FOR STATIC PUSHOVER, CYCLIC DEGRADATION, STATIC TIME-HISTORY AND DYNAMIC TIME-HISTORY ANALYSIS

THIS PROGRAM IS WRITTEN BY SALAR DELAVAR GHASHGHAEE (QASHQAI)



$$\text{Structural Ductility Damage Index} = \frac{\Delta_d - \Delta_y}{\Delta_u - \Delta_y}$$

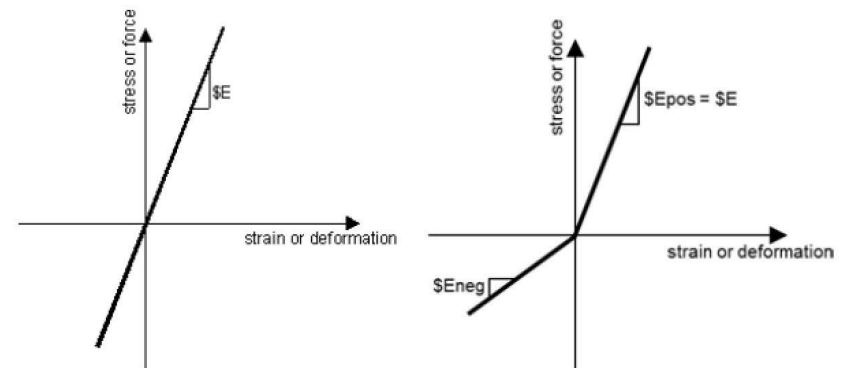
$\Delta_d$  = Lateral Displaement from Dynamic Analysis

$\Delta_y$  = Lateral Yield Displaement from Pushover Analysis

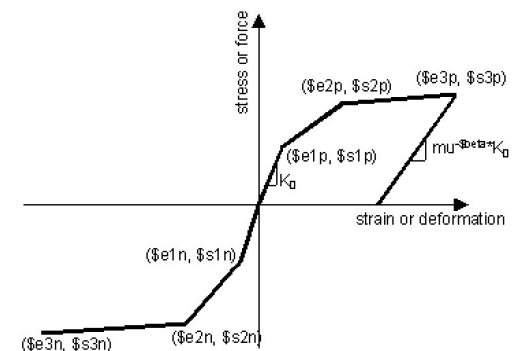
$\Delta_u$  = Lateral Ultimate Displaement from Pushover Analysis

$$P(t) = P_0 e^{-0.05\bar{\omega}t} \sin(\bar{\omega}t)$$

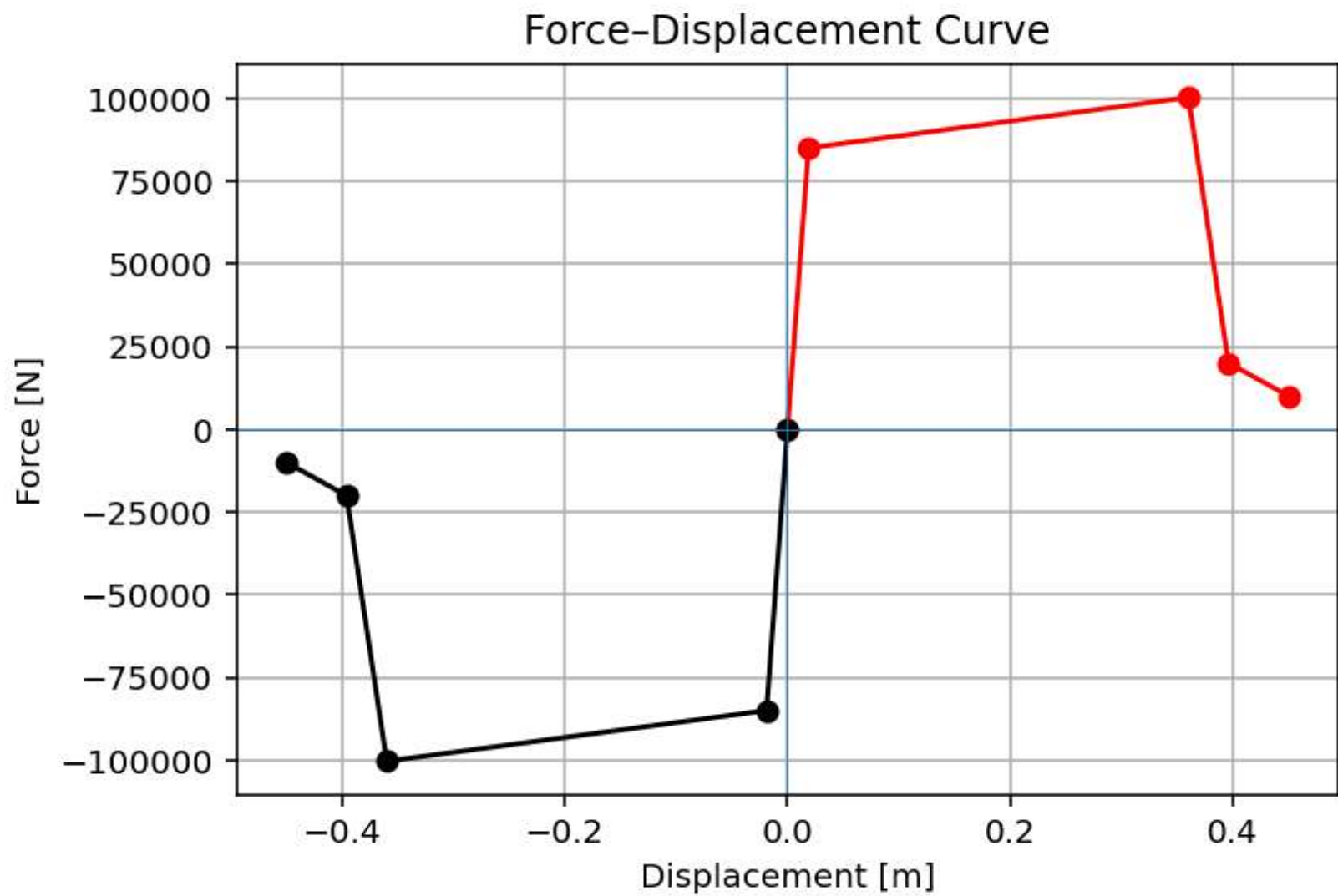
$$m\ddot{u} + c\dot{u} + ku = P(t)$$



**ELASTIC MATERIAL**



**INELASTIC MATERIAL**



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C:\Users\De\l\Desktop\OPENSEES\_FILES\+EIGHT\_ANALY...ION\_CAPACITY\_INDEX\+SDOF\_8\_ANA\p(t)\_SDOF\_8\_ANA.py

P(t)\_SDOF\_8\_ANA.py

```
1 #####
2 # >> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<
3 # COMPREHENSIVE NONLINEAR SEISMIC ASSESSMENT OF A SINGLE-DEGREE-FREEDOM STRUCTURE : AN OPE
4 # FOR STATIC PUSHOVER, CYCLIC DEGRADATION, STATIC TIME-HISTORY AND DYNAMIC TIME-HISTORY
5 #
6 # P(t) = P0 sin(wt)
7 # P(t) = P0 exp(-0.05wt) sin(wt)
8 #
9 # ASSESSMENT OF DUCTILITY DAMAGE INDICES FOR STRUCTURAL ELEMENTS AND SYSTEMS AND E
10 # OF ENERGY DISSIPATION CAPACITY INDEX
11 #
12 # THIS PROGRAM WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)
13 # EMAIL: salar.d.ghashghaei@gmail.com
14 #####
15 """
16 Nonlinear Seismic Performance Assessment of a SDOF:
17 An OpenSeesPy Framework for Material and Geometric Nonlinearity Under Static, Cyclic, and
18
19 This OpenSeesPy script performs rigorous nonlinear static and dynamic analysis of a SDOF
20 for performance-based earthquake engineering. The 2D model incorporates both material non
21 (elastic-perfectly plastic or hysteretic steel with strain hardening)
22 and geometric nonlinearity (corotational truss formulation) to capture P-delta effects
23 and large displacements-essential for collapse assessment.
24
25 Eight analysis protocols are implemented:
26 (1) [PERIOD] : Structural Period
27 (2) [STATIC] : Gravity load analysis establishing dead load state
28 (3) [PUSHOVER] : Displacement-controlled pushover generating full capacity curves
29 and plastic mechanism identification
30 (4) [CYCLIC_DISPLACEMENT] : Symmetric cyclic displacement protocol capturing hysteresis,
31 pinching behavior, and energy dissipation degradation
32 (5) [STATIC_EXTERNAL_TIME-DEPENDENT_LOADING] : Static Analysis of External time-dependent
33 (6) [DYNAMIC_EXTERNAL_TIME-DEPENDENT_LOADING] : Dynamic Analysis of External time-dependen
34 (7) [FREE-VIBRATION] : Free-vibration with initial conditions extracting damping ratios
```

30 %

Force-Displacement Curve

| Displacement [m] | Force [N] | Curve Type       |
|------------------|-----------|------------------|
| -0.4             | -100000   | Black Hysteresis |
| -0.4             | -25000    | Black Hysteresis |
| 0.0              | 0         | Black Hysteresis |
| 0.0              | 0         | Red Curve        |
| 0.4              | 100000    | Red Curve        |
| 0.4              | 25000     | Red Curve        |
| 0.0              | 0         | Red Curve        |

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# STATIC ANALYSIS

Spyder (Python 3.12)

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P(t)\_SDOF\_8\_ANA.py

```
755 # STATIC ANALYSIS
756 MAT_TYPE = 'INELASTIC' # 'ELASTIC' OR 'INELASTIC'
757 ANAL_TYPE = 'STATIC'
758
759 DATA = SDOF(MAT_TYPE, TOTAL_MASS)
760 (reaction, disp) = DATA
761
762 #####
763 # PUSHOVER ANALYSIS (STATIC TIME-HISTORY ANALYSIS)
764 MAT_TYPE = 'INELASTIC' # 'ELASTIC' OR 'INELASTIC'
765 ANAL_TYPE = 'PUSHOVER'
766
767 DATA = SDOF(MAT_TYPE, TOTAL_MASS)
768 (reaction_PUSH, disp_PUSH,
769  PERIOD_MIN_PUSH, PERIOD_MAX_PUSH) = DATA
770
771
772 XDATA = disp_PUSH
773 YDATA = reaction_PUSH
774 XLABEL = 'Displacement [m]'
775 YLABEL = 'Base Reaction [N]'
776 TITLE = 'Base Reaction and Displacement of Structure During Pushover Analysis'
777 COLOR = 'black'
778 SEMILOGY = False
779 PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMILOGY)
780
781 DATA = S07.BILINEAR_CURVE(np.abs(disp_PUSH), np.abs(reaction_PUSH), SLOPE_NODE=10)
782 (X_PUSH, Y_PUSH, Elastic_ST, Plastic_ST, Tangent_ST, Ductility_Rito, Over_Strength_Factor)
783
784 # PLOT STRUCTURAL PERIOD DURING THE ANALYSIS
785 plt.figure(0, figsize=(12, 8))
786 plt.plot(disp_PUSH, PERIOD_MIN_PUSH, linewidth=3)
787 plt.plot(disp_PUSH, PERIOD_MAX_PUSH, linewidth=3)
788 plt.title('Period of Structure During Pushover Analysis')
```

Force-Displacement Curve

| Displacement [m] | Force [N] (Black Curve) | Force [N] (Red Curve) |
|------------------|-------------------------|-----------------------|
| -0.4             | -100000                 | -                     |
| -0.2             | -75000                  | -                     |
| 0.0              | 0                       | 0                     |
| 0.2              | 75000                   | -                     |
| 0.4              | 100000                  | 100000                |
| 0.4              | -                       | 25000                 |

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# **PUSHOVER ANALYSIS**



Spyder (Python 3.12)

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P(t)\_SDOF\_8\_ANA.py

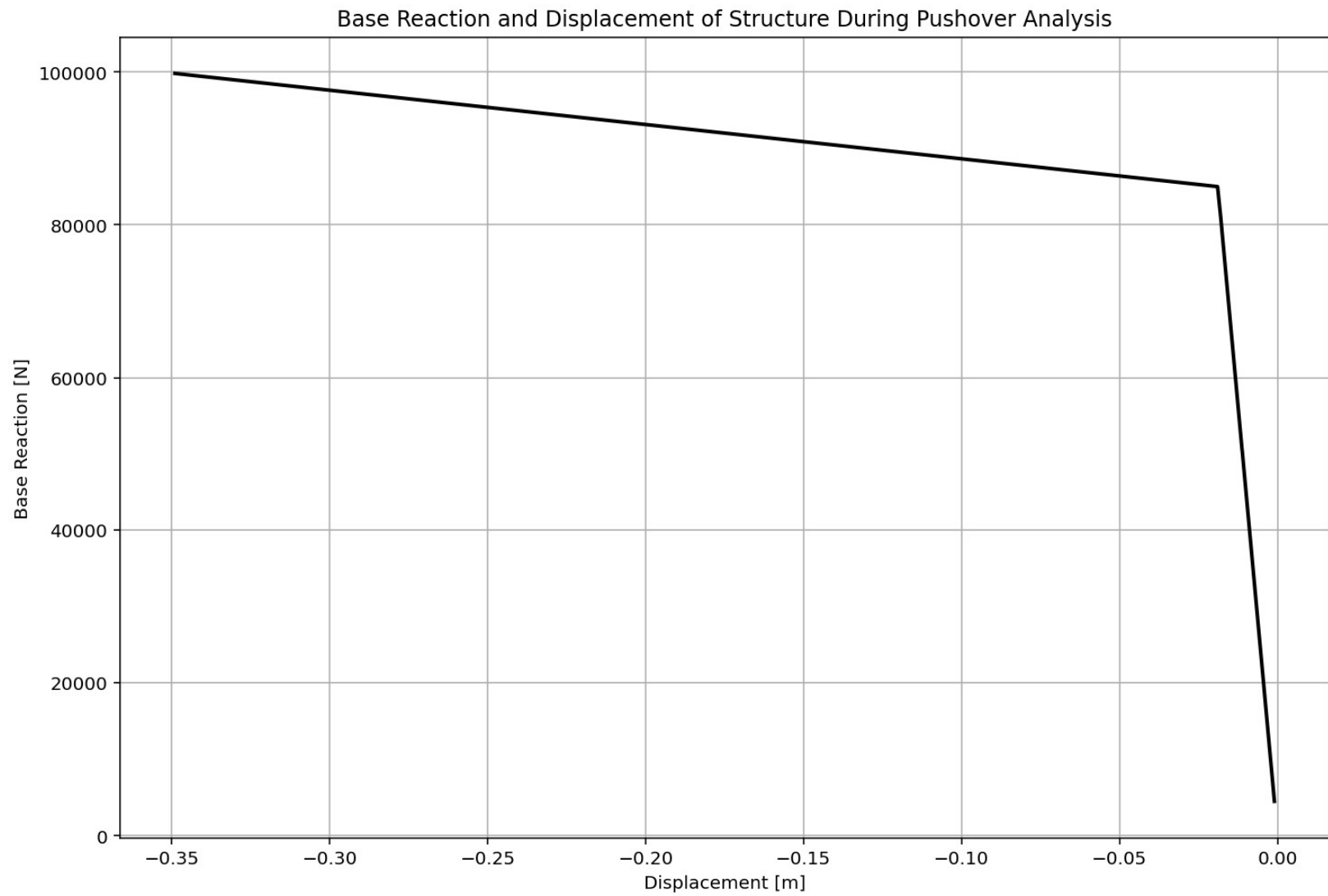
```
763 # PUSHOVER ANALYSIS (STATIC TIME-HISTORY ANALYSIS)
764 MAT_TYPE = 'INELASTIC' # 'ELASTIC' OR 'INELASTIC'
765 ANAL_TYPE = 'PUSHOVER'
766
767 DATA = SDOF(MAT_TYPE, TOTAL_MASS)
768 (reaction_PUSH, disp_PUSH,
769  PERIOD_MIN_PUSH, PERIOD_MAX_PUSH) = DATA
770
771
772 XDATA = disp_PUSH
773 YDATA = reaction_PUSH
774 XLABEL = 'Displacement [m]'
775 YLABEL = 'Base Reaction [N]'
776 TITLE = 'Base Reaction and Displacement of Structure During Pushover Analysis'
777 COLOR = 'black'
778 SEMIOLOGY = False
779 PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMIOLOGY)
780
781 DATA = S07.BILNEAR_CURVE(np.abs(disp_PUSH), np.abs(reaction_PUSH), SLOPE_NODE=10)
782 (X_PUSH, Y_PUSH, Elastic_ST, Plastic_ST, Tangent_ST, Ductility_Rito, Over_Strength_Factor)
783
784 # PLOT STRUCTURAL PERIOD DURING THE ANALYSIS
785 plt.figure(0, figsize=(12, 8))
786 plt.plot(disp_PUSH, PERIOD_MIN_PUSH, linewidth=3)
787 plt.plot(disp_PUSH, PERIOD_MAX_PUSH, linewidth=3)
788 plt.title('Period of Structure During Pushover Analysis')
789 plt.ylabel('Structural Period [s]')
790 plt.xlabel('Displacement [m]')
791 #plt.semilogy()
792 plt.grid()
793 plt.legend([f'PERIOD - MIN VALUES: Min: {np.min(PERIOD_MIN_PUSH):.3f} (s) - Mean: {np.mean(PERIOD_MIN_PUSH):.3f} (s)',
794            f'PERIOD - MAX VALUES: Min: {np.min(PERIOD_MAX_PUSH):.3f} (s) - Mean: {np.mean(PERIOD_MAX_PUSH):.3f} (s)'])
795
796 plt.show()
```

Base Reaction and Displacement of Structure During Pushover Analysis

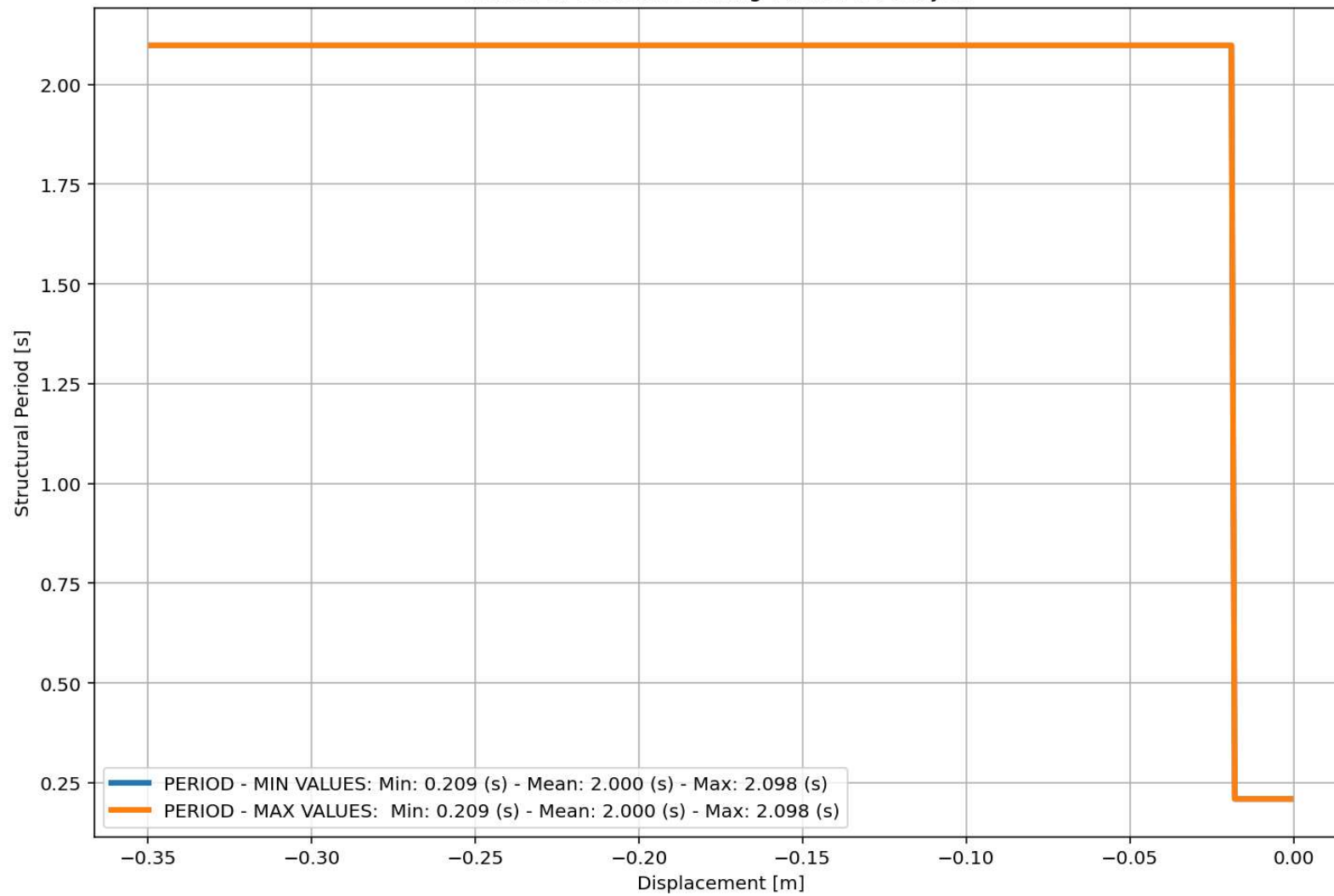
Python Console Files Help Variable Explorer Debugger History Plots

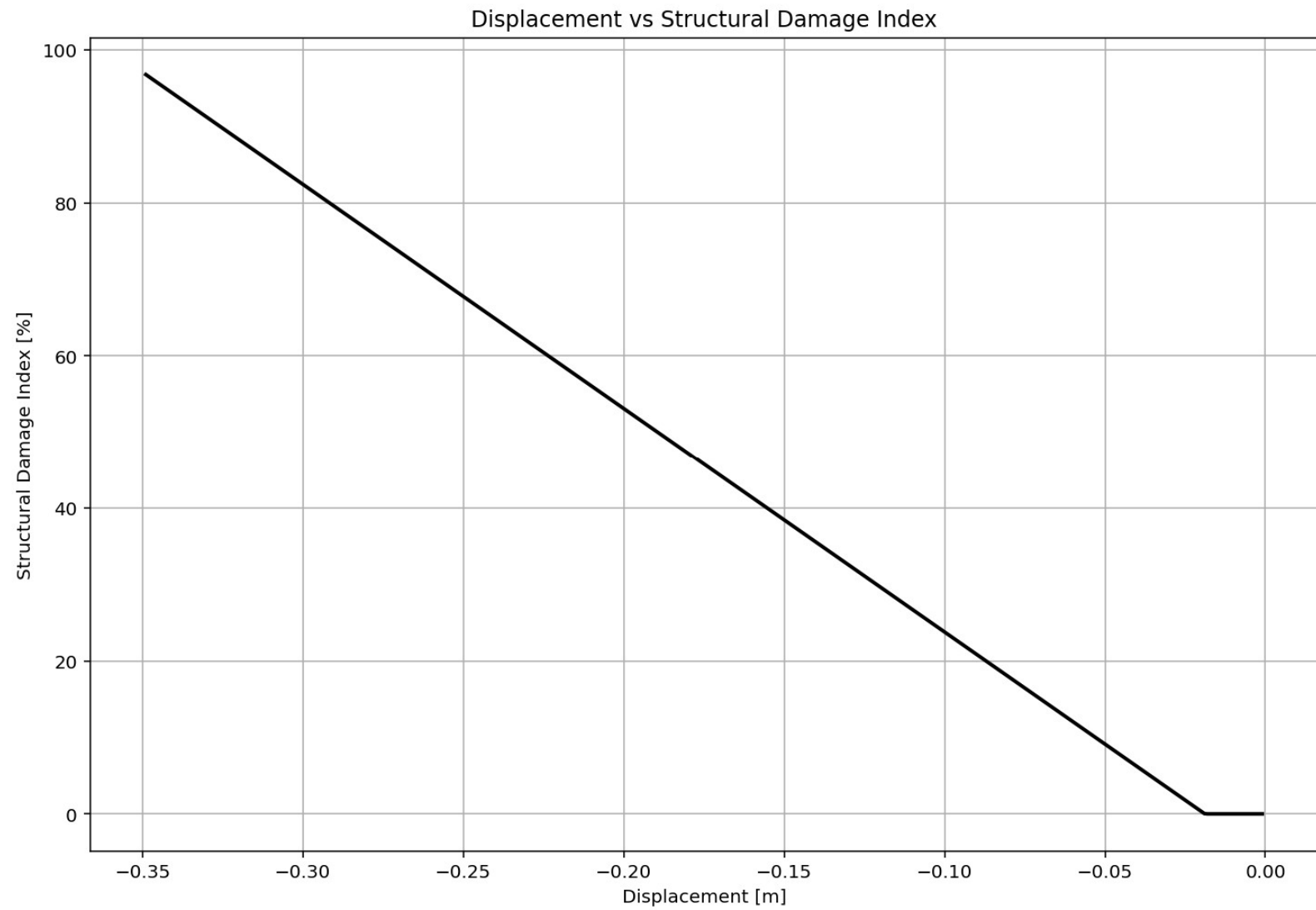
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Period of Structure During Pushover Analysis





# **CYCLIC DISPLACEMENT ANALYSIS**

Spyder (Python 3.12)

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P(t)\_SDOF\_8\_ANA.py

```
800 # CYCLIC DISPLACEMENT ANALYSIS (STATIC TIME-HISTORY ANALYSIS)
801 MAT_TYPE = 'INELASTIC' # 'ELASTIC' OR 'INELASTIC'
802 ANAL_TYPE = 'CYCLIC_DISPLACEMENT'
803
804 DATA = SDOF(MAT_TYPE, TOTAL_MASS)
805 (reaction_CP, disp_CP,
806  PERIOD_MIN_CP, PERIOD_MAX_CP) = DATA
807
808
809 XDATA = disp_CP
810 YDATA = reaction_CP
811 XLABEL = 'Displacement [m]'
812 YLABEL = 'Base Reaction [N]'
813 TITLE = 'Base Reaction and Displacement of Structure During Cyclic-Displacement Analysis'
814 COLOR = 'black'
815 SEMILOGY = False
816 PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMILOGY)
817
818 # PLOT STRUCTURAL PERIOD DURING THE ANALYSIS
819 plt.figure(0, figsize=(12, 8))
820 plt.plot(disp_CP, PERIOD_MIN_CP, linewidth=3)
821 plt.plot(disp_CP, PERIOD_MAX_CP, linewidth=3)
822 plt.title('Period of Structure During Cyclic Displacement Analysis')
823 plt.ylabel('Structural Period [s]')
824 plt.xlabel('Displacement [m]')
825 #plt.semilogy()
826 plt.grid()
827 plt.legend([f'PERIOD - MIN VALUES: Min: {np.min(PERIOD_MIN_CP):.3f} (s) - Mean: {np.mean(P
828 f'PERIOD - MAX VALUES: Min: {np.min(PERIOD_MAX_CP):.3f} (s) - Mean: {np.mean(
829 ]})
830 plt.show()
831
832
833 #%%-----
```

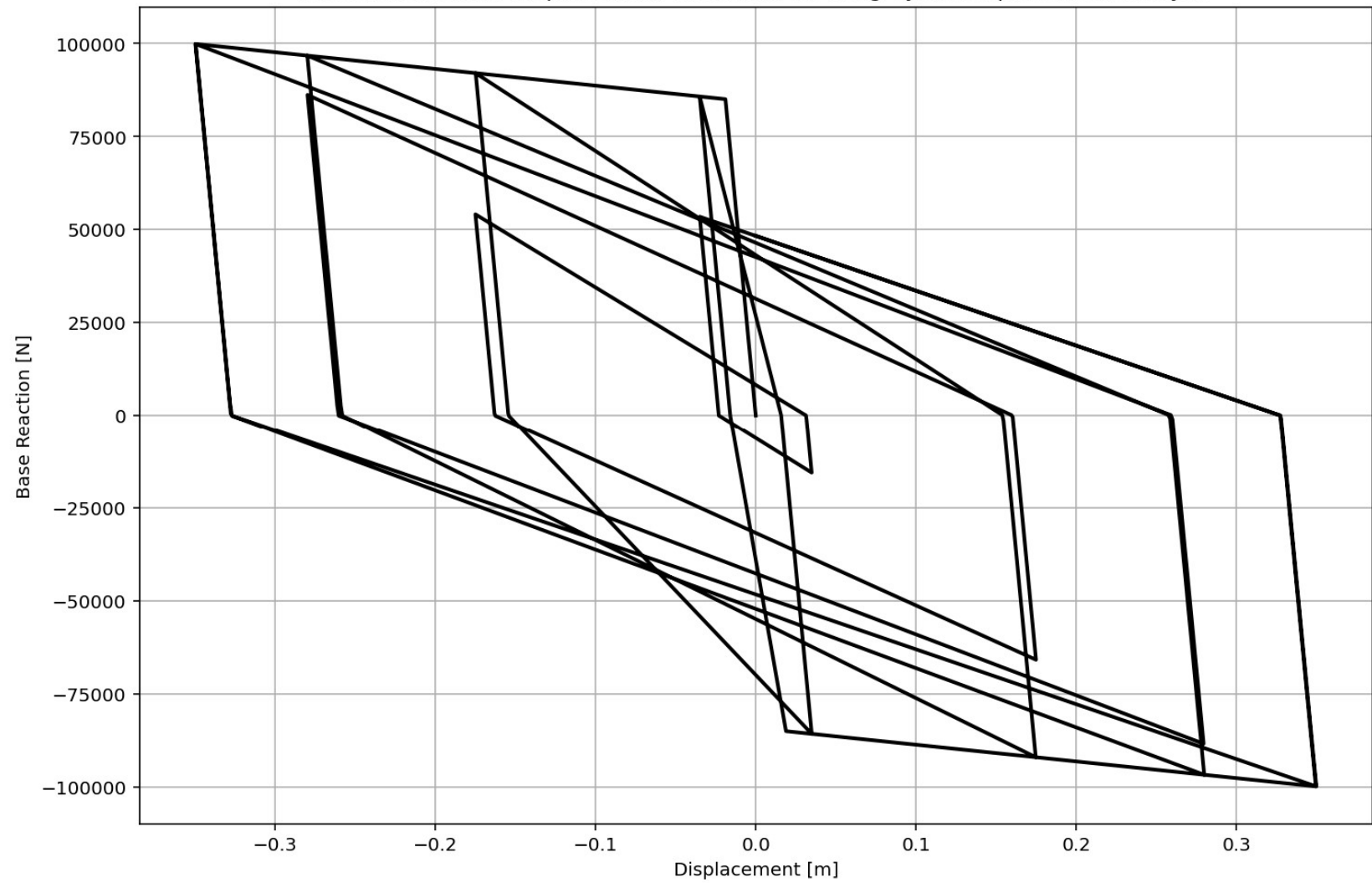
Base Reaction and Displacement of Structure During Cyclic-Displacement Analysis

30 %

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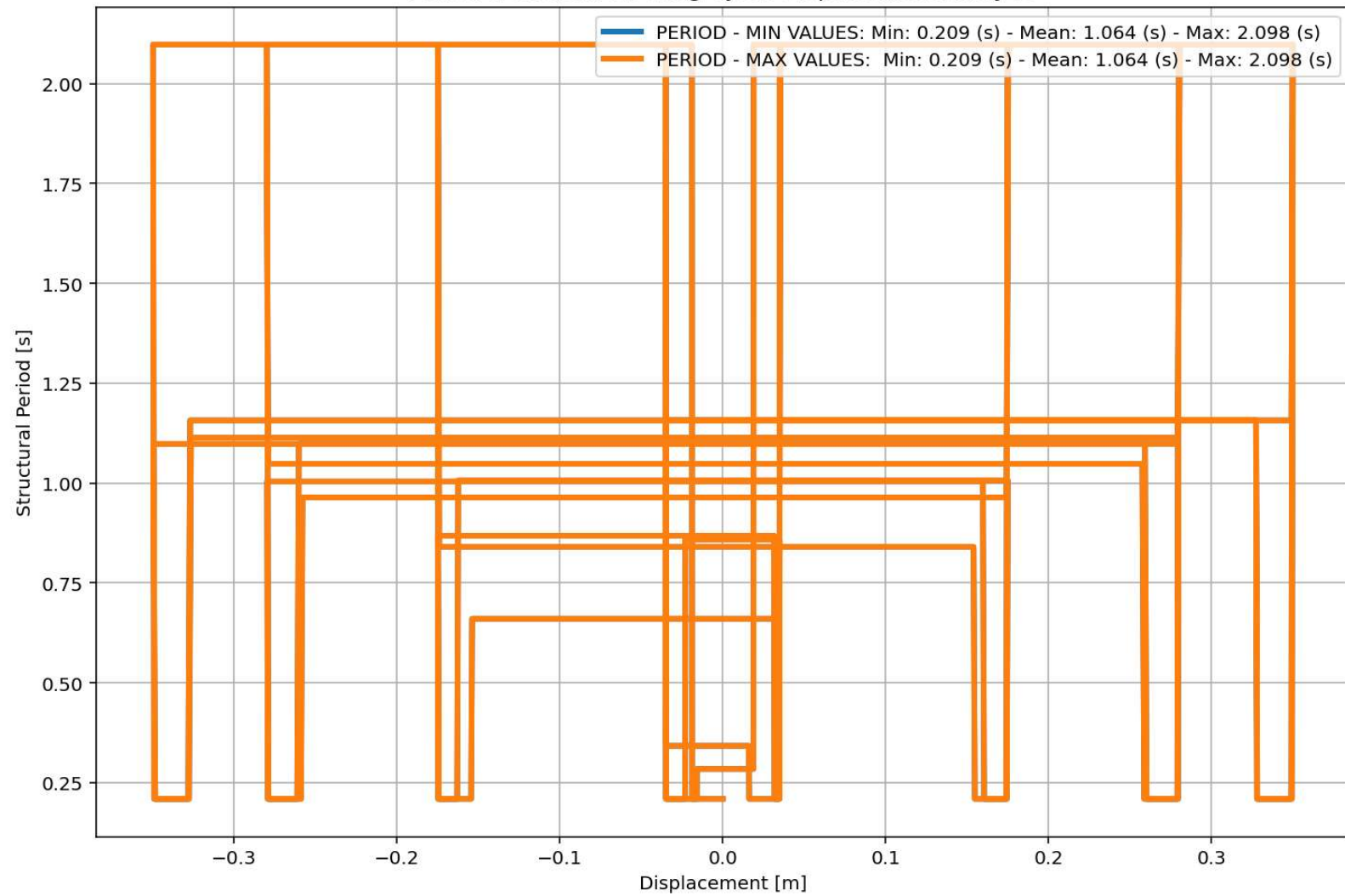
Inline Conda: anaconda3 (Python 3.12.7) ✓ LSP: Python Line 9, Col 86 UTF-8 CRLF RW Mem 42%

Base Reaction and Displacement of Structure During Cyclic-Displacement Analysis



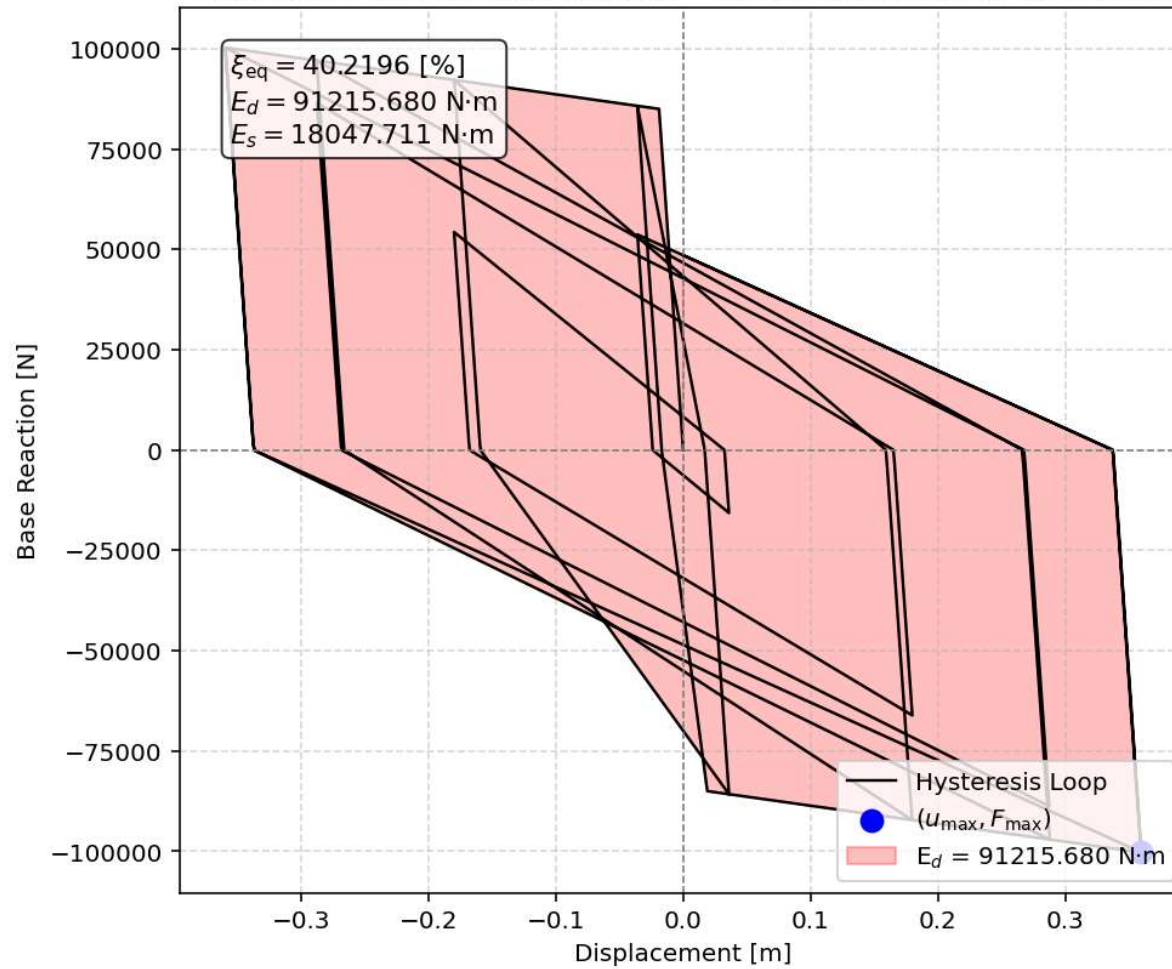


Period of Structure During Cyclic Displacement Analysis





Equivalent viscous damping ratio - Cyclic Displacement Hysteresis



# **STATIC TIME-HISTORY WITH EXTERNAL TIME-DEPENDENT LOADING ANALYSIS**

Spyder (Python 3.12)

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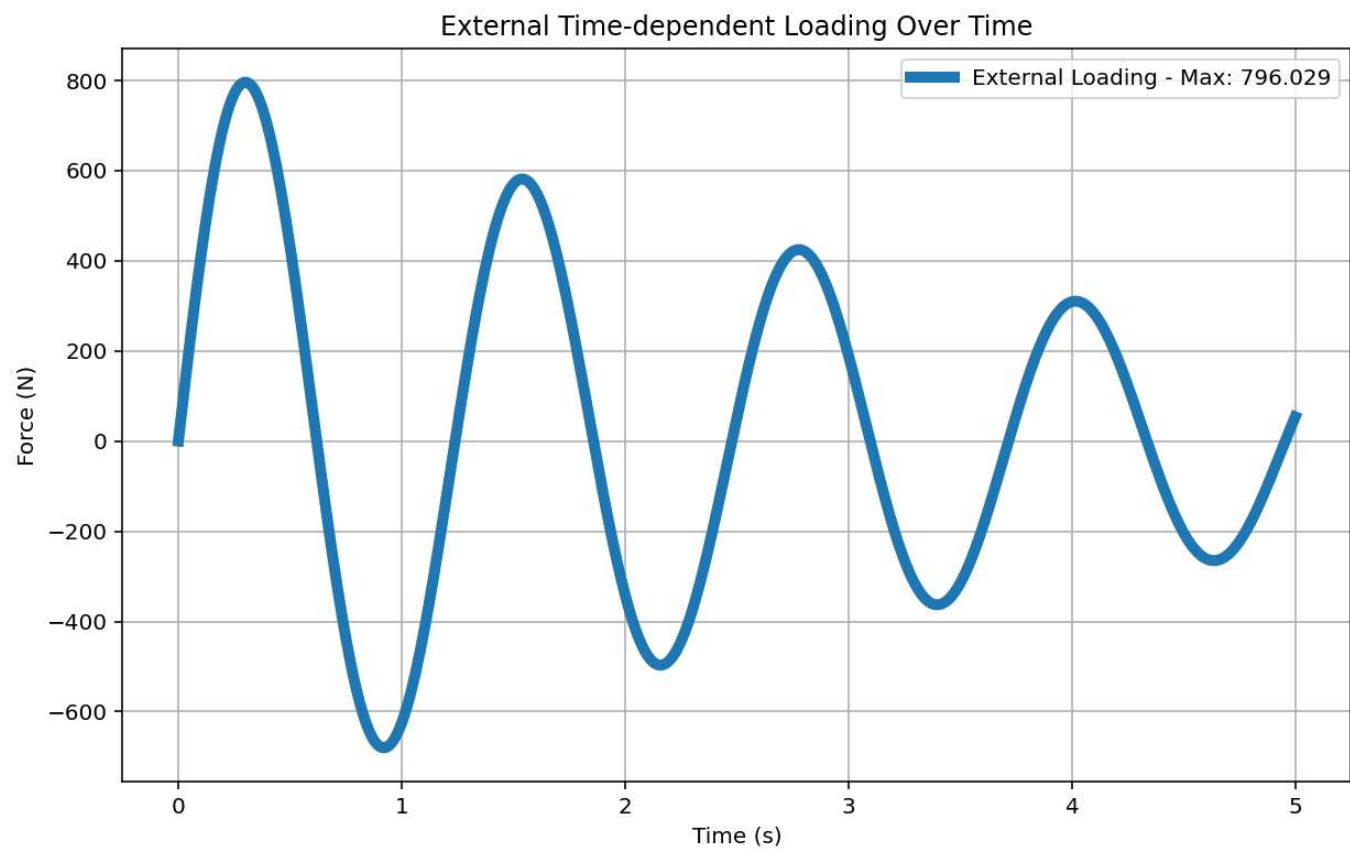
P(t)\_SDOF\_8\_ANA.py

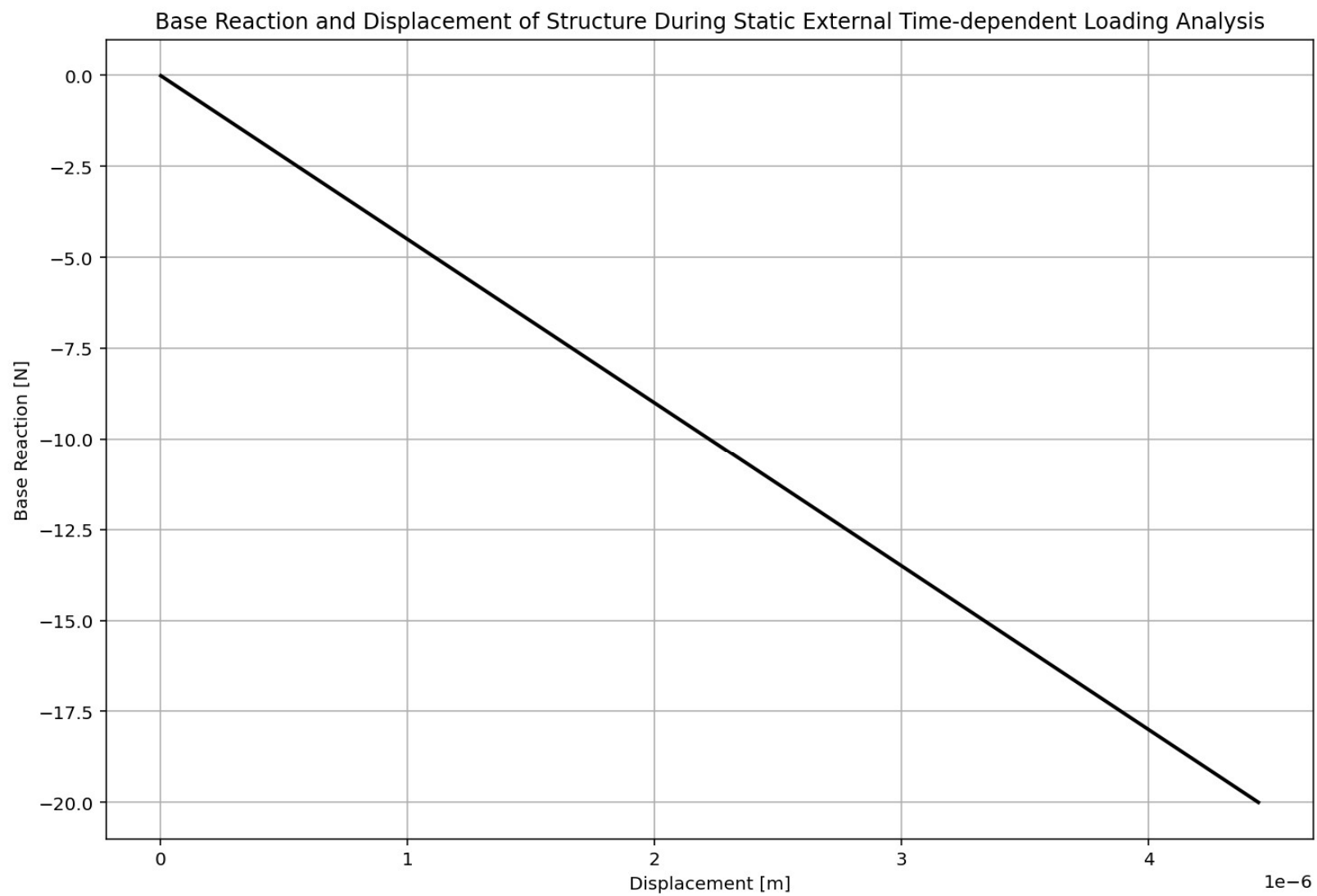
```
834 # EXTERNAL TIME-DEPENDENT LOADING ANALYSIS (STATIC TIME-HISTORY ANALYSIS)
835 MAT_TYPE = 'INELASTIC' # 'ELASTIC' OR 'INELASTIC'
836 ANAL_TYPE = 'STATIC_EXTERNAL_TIME-DEPENDENT_LOADING'
837
838 DATA = SDOF(MAT_TYPE, TOTAL_MASS)
839
840 (reaction_ETDLS, disp_ETDLS,
841  PERIOD_MIN_ETDLS, PERIOD_MAX_ETDLS) = DATA
842
843
844 XDATA = disp_ETDLS
845 YDATA = reaction_ETDLS
846 XLABEL = 'Displacement [m]'
847 YLABEL = 'Base Reaction [N]'
848 TITLE = 'Base Reaction and Displacement of Structure During Static External Time-dependent
849         Loading Analysis'
850 COLOR = 'black'
851 SEMIOLOGY = False
852 PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMIOLOGY)
853
854 # PLOT STRUCTURAL PERIOD DURING THE ANALYSIS
855 plt.figure(0, figsize=(12, 8))
856 plt.plot(disp_ETDLS, PERIOD_MIN_ETDLS, linewidth=3)
857 plt.plot(disp_ETDLS, PERIOD_MAX_ETDLS, linewidth=3)
858 plt.title('Period of Structure During Static External Time-dependent Loading Analysis')
859 plt.ylabel('Structural Period [s]')
860 plt.xlabel('Displacement [m]')
861 #plt.semilogy()
862 plt.grid()
863 plt.legend([f'PERIOD - MIN VALUES: Min: {np.min(PERIOD_MIN_ETDLS):.3f} (s) - Mean: {np.mean(PERIOD_MIN_ETDLS):.3f} (s)',
864            f'PERIOD - MAX VALUES: Min: {np.min(PERIOD_MAX_ETDLS):.3f} (s) - Mean: {np.mean(PERIOD_MAX_ETDLS):.3f} (s)'],
865            loc='best')
866 plt.show()
867
```

Base Reaction and Displacement of Structure During Static External Time-dependent Loading Analysis

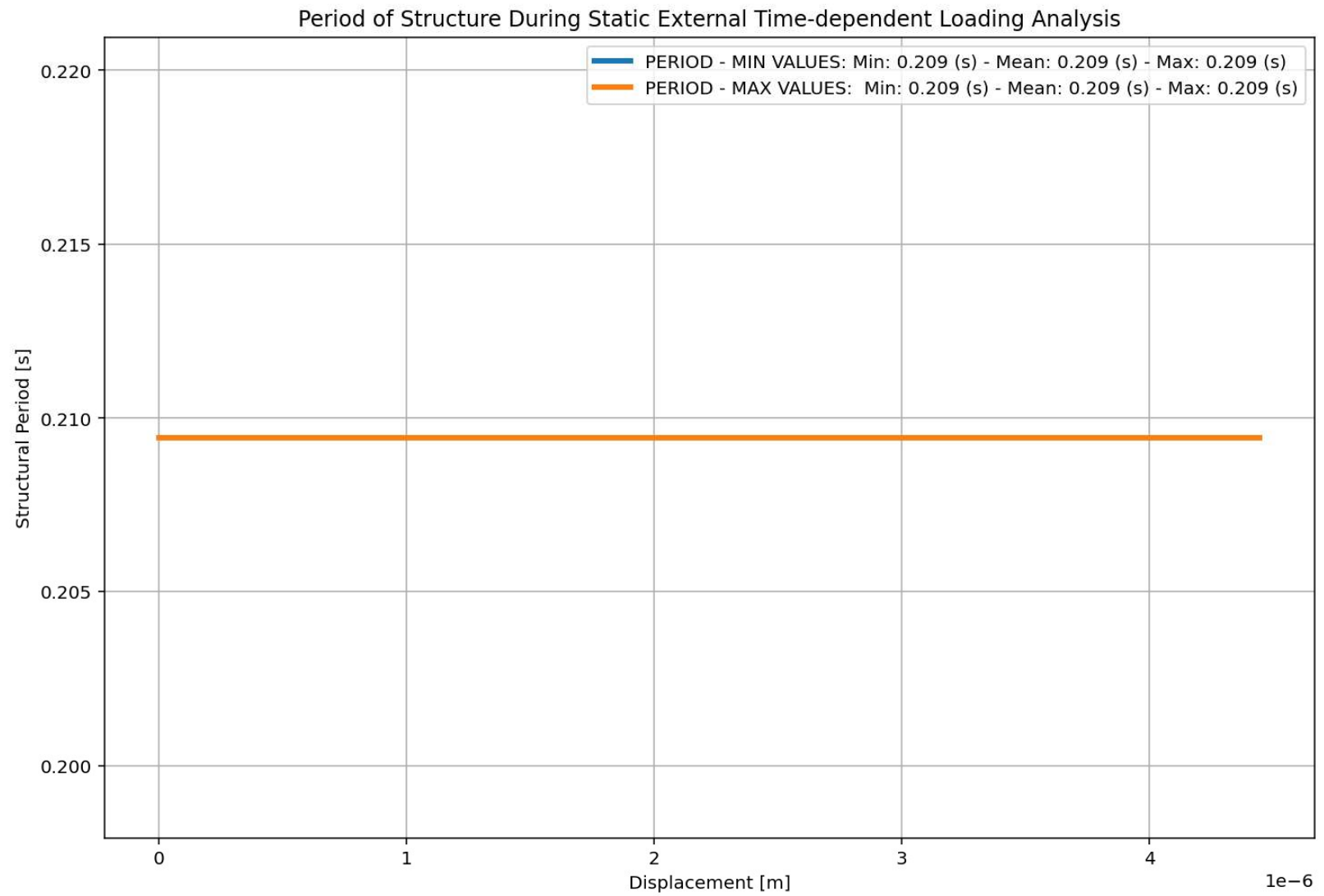
Python Console Files Help Variable Explorer Debugger History Plots

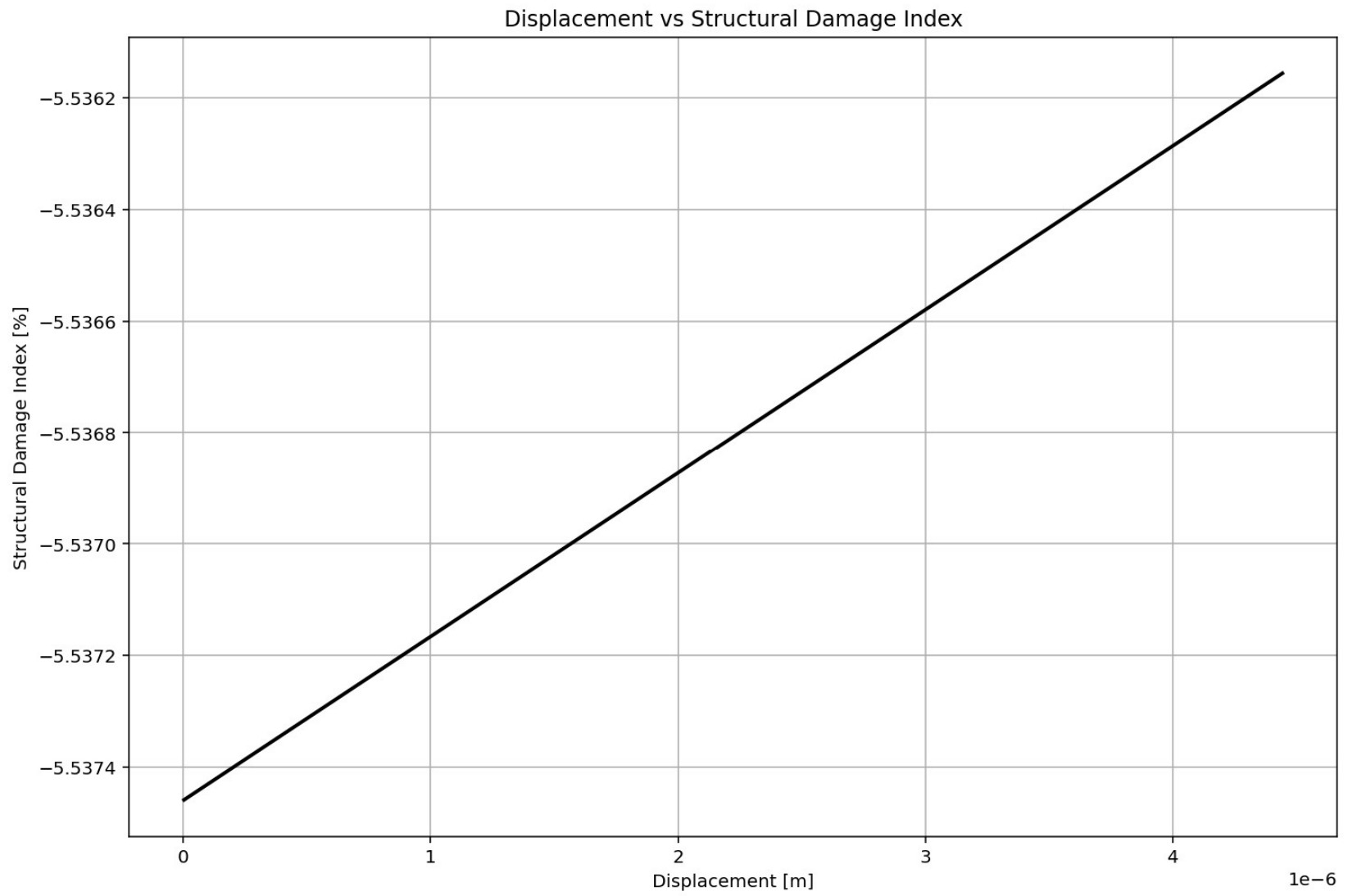
Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 855, Col 35 UTF-8 CRLF RW Mem 42%











# **DYNAMIC TIME-HISTORY WITH EXTERNAL TIME-DEPENDENT LOADING ANALYSIS**

Spyder (Python 3.12)

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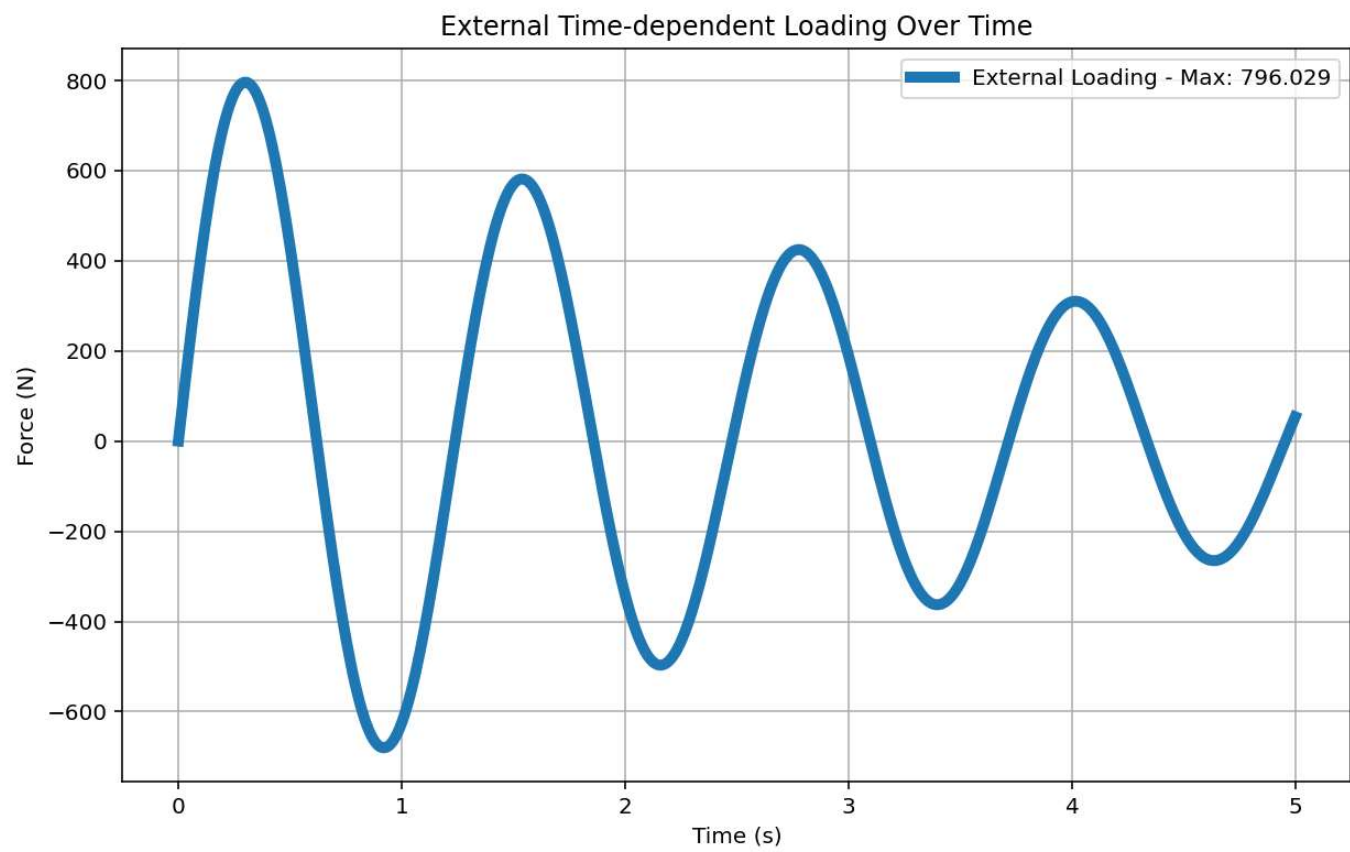
P(t)\_SDOF\_8\_ANA.py

```
869 # EXTERNAL TIME-DEPENDENT LOADING ANALYSIS (DYNAMIC TIME-HISTORY ANALYSIS)
870 #ELE_TYPE = 'Truss' # MATERIAL NONLINEARITY
871 ELE_TYPE = 'corotTruss' # MATERIAL AND GEOMETRIC NONLINEARITIES
872 MAT_TYPE = 'INELASTIC' # 'ELASTIC' OR 'INELASTIC'
873 ANAL_TYPE = 'DYNAMIC_EXTERNAL_TIME-DEPENDENT_LOADING'
874
875 DATA = SDOF(MAT_TYPE, TOTAL_MASS)
876
877 (time_ETDLD, reaction_ETDLD, disp_ETDLD, velo_ETDLD, acc_ETDLD,
878 stiffness, PERIOD, damping_ratio,
879 PERIOD_MIN_ETDLD, PERIOD_MAX_ETDLD) = DATA
880
881
882
883 XDATA = disp_ETDLD
884 YDATA = reaction_ETDLD
885 XLABEL = 'Displacement [m]'
886 YLABEL = 'Base Reaction [N]'
887 TITLE = 'Base Reaction and Displacement of Structure During Dynamic External Time-dependen
888 COLOR = 'black'
889 SEMILOGY = False
890 PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMILOGY)
891
892 PLOT_TIME_HISTORY(time_ETDLD, reaction_ETDLD, disp_ETDLD, velo_ETDLD, acc_ETDLD)
893
894 # PLOT STRUCTURAL PERIOD DURING THE ANALYSIS
895 plt.figure(0, figsize=(12, 8))
896 plt.plot(disp_ETDLD, PERIOD_MIN_ETDLD, linewidth=3)
897 plt.plot(disp_ETDLD, PERIOD_MAX_ETDLD, linewidth=3)
898 plt.title('Period of Structure During Dynamic External Time-dependent Loading Analysis')
899 plt.ylabel('Structural Period [s]')
900 plt.xlabel('Displacement [m]')
901 #plt.semilogy()
902 plt.grid()
```

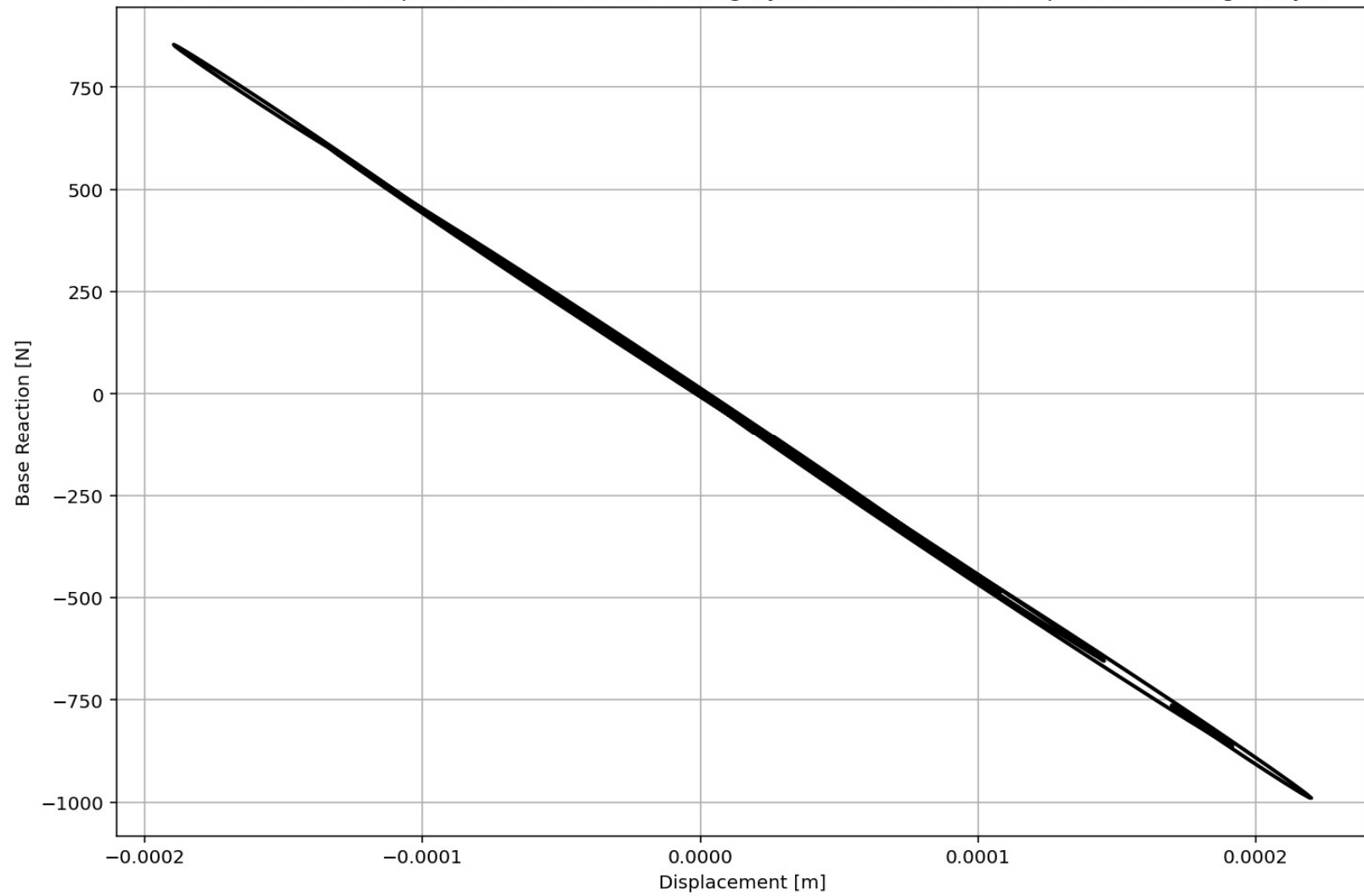
Base Reaction and Displacement of Structure During Dynamic External Time-dependent Loading Analysis

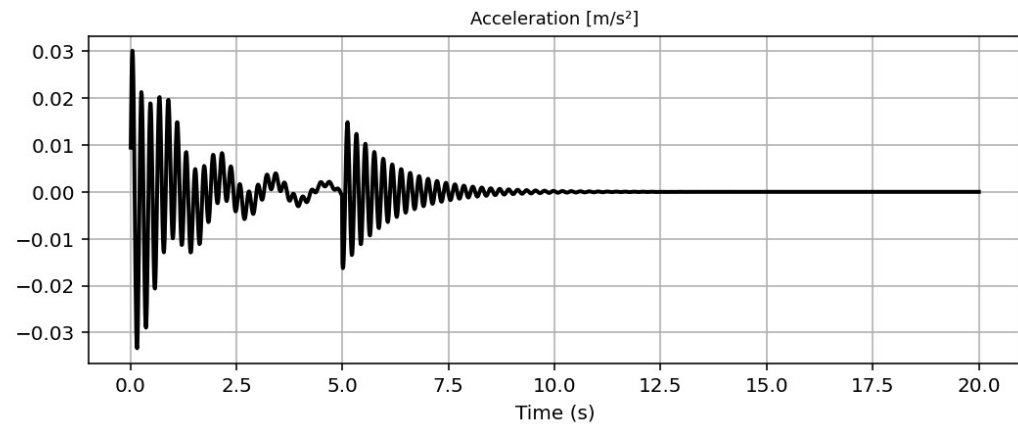
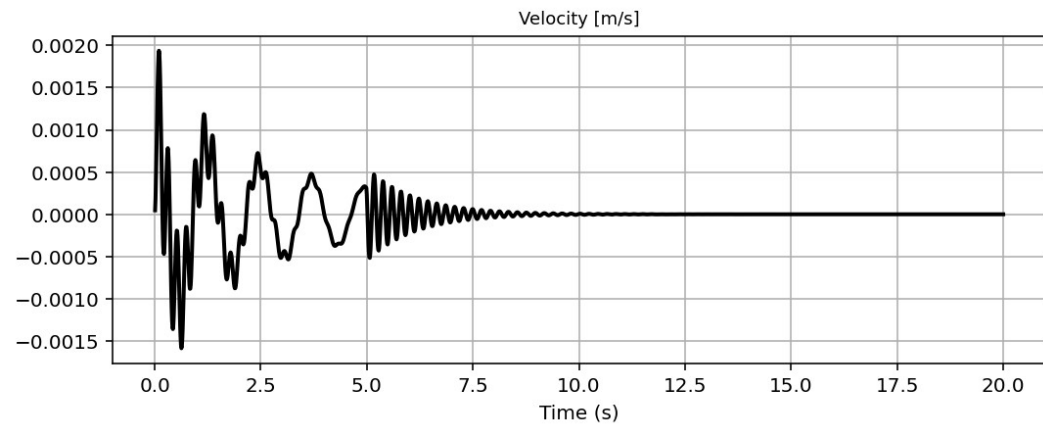
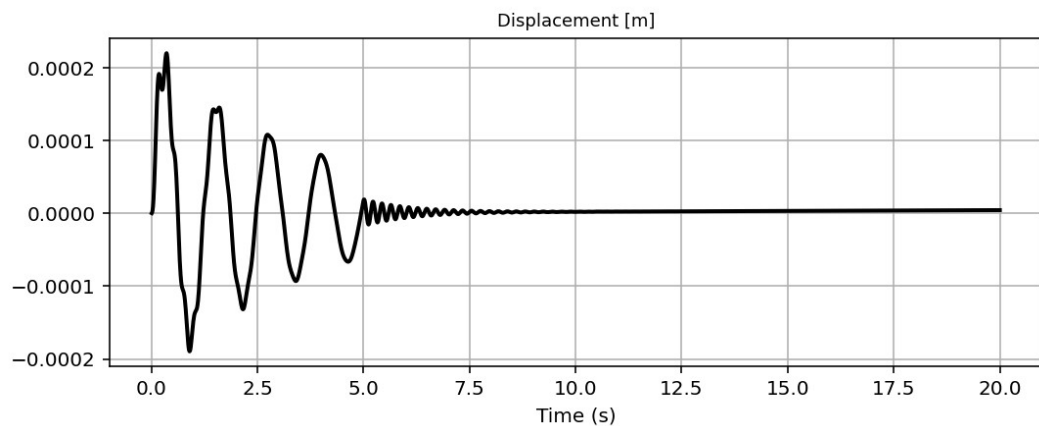
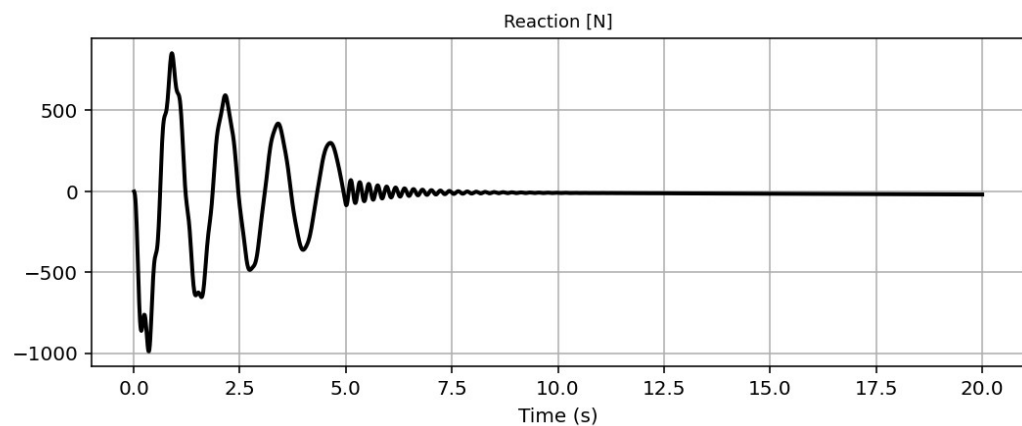
Python Console Files Help Variable Explorer Debugger History Plots

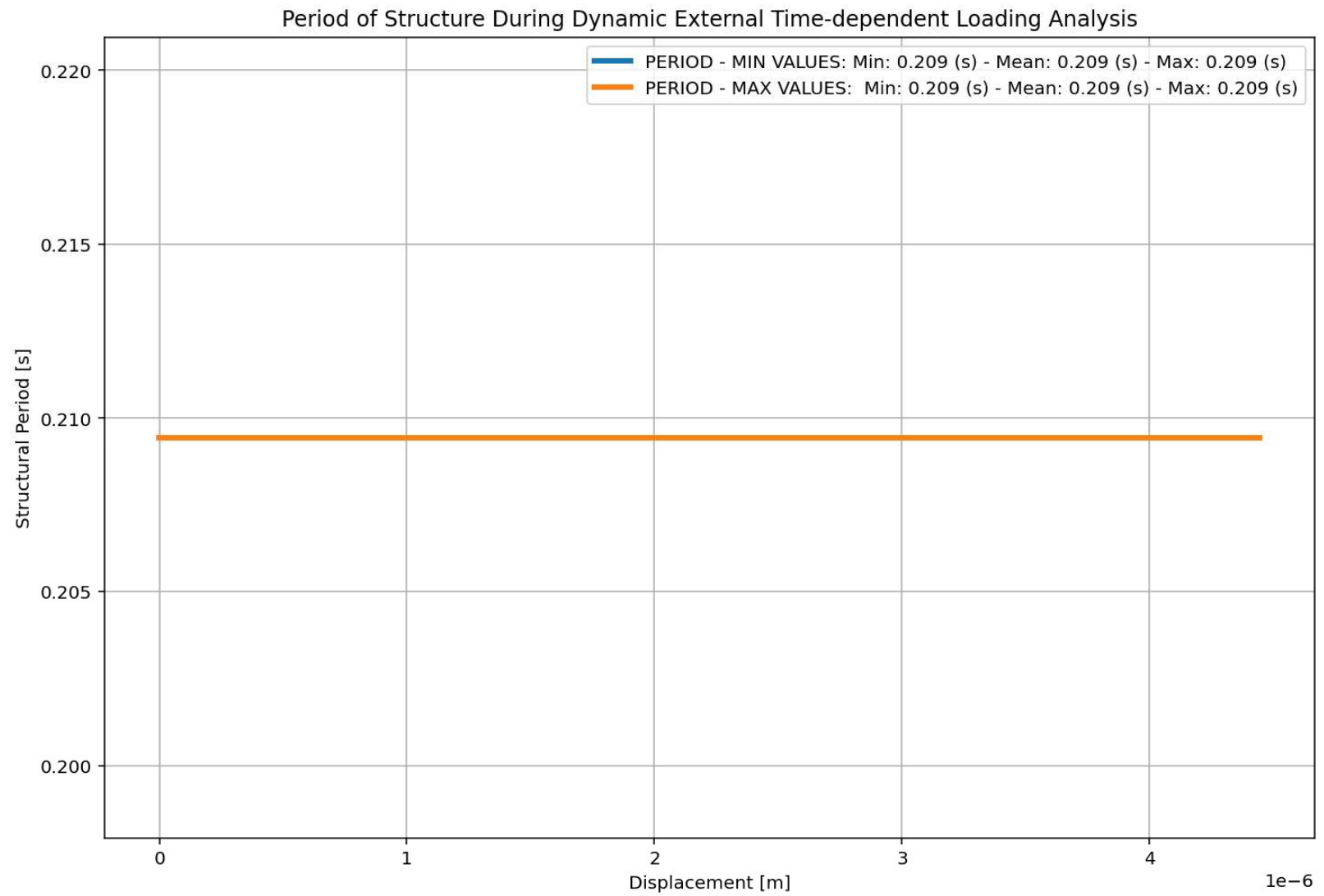
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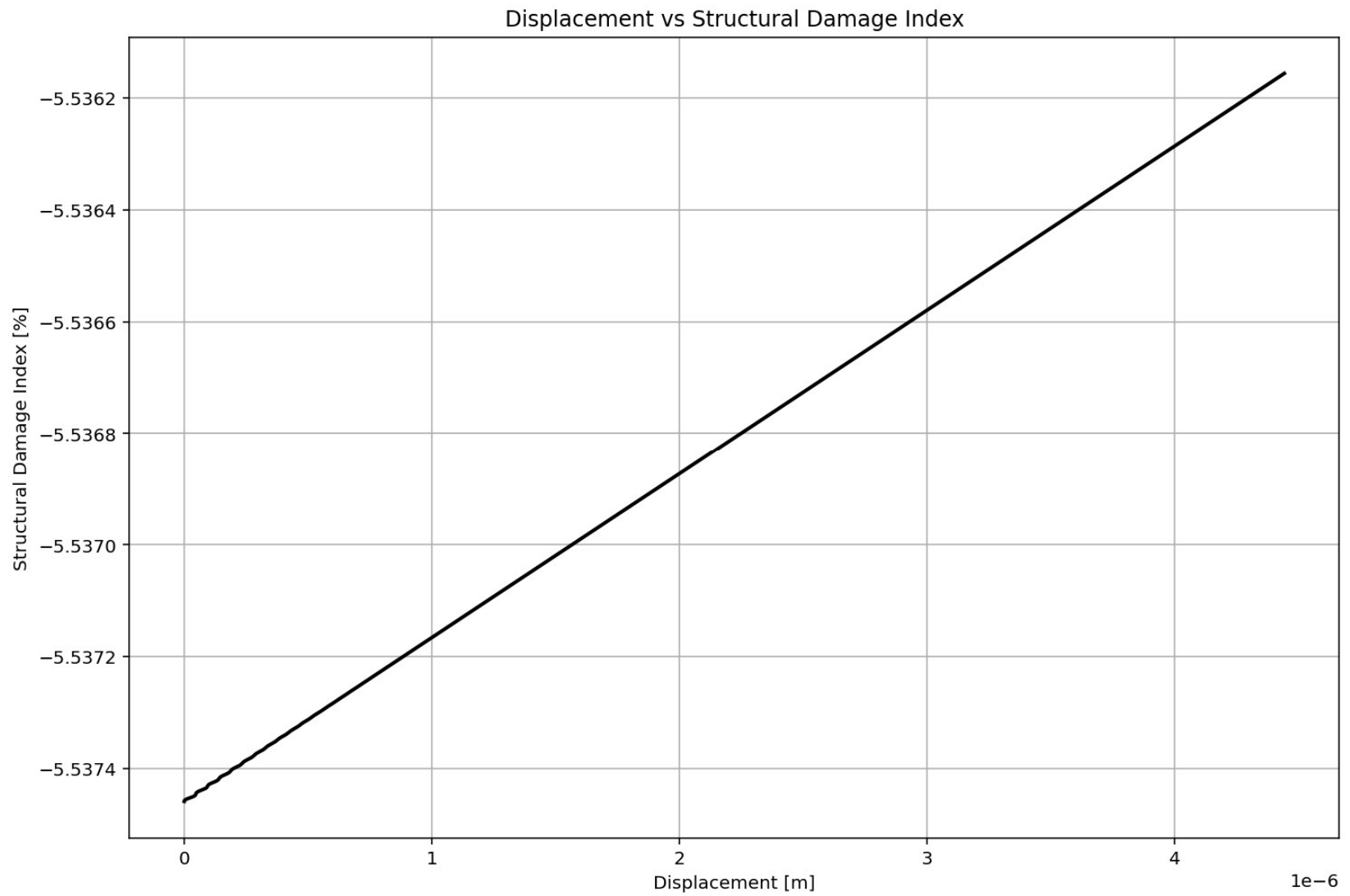
Base Reaction and Displacement of Structure During Dynamic External Time-dependent Loading Analysis



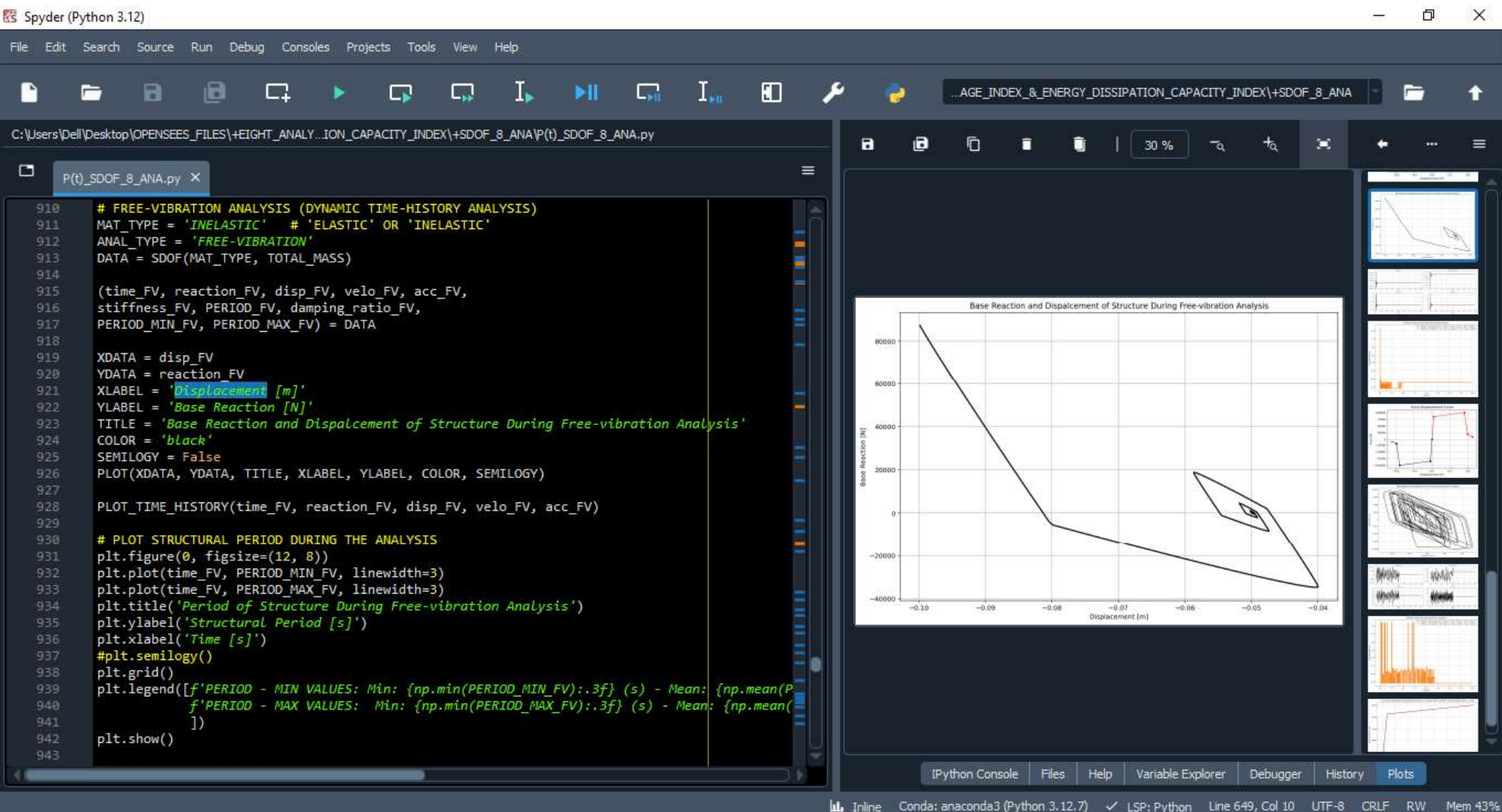




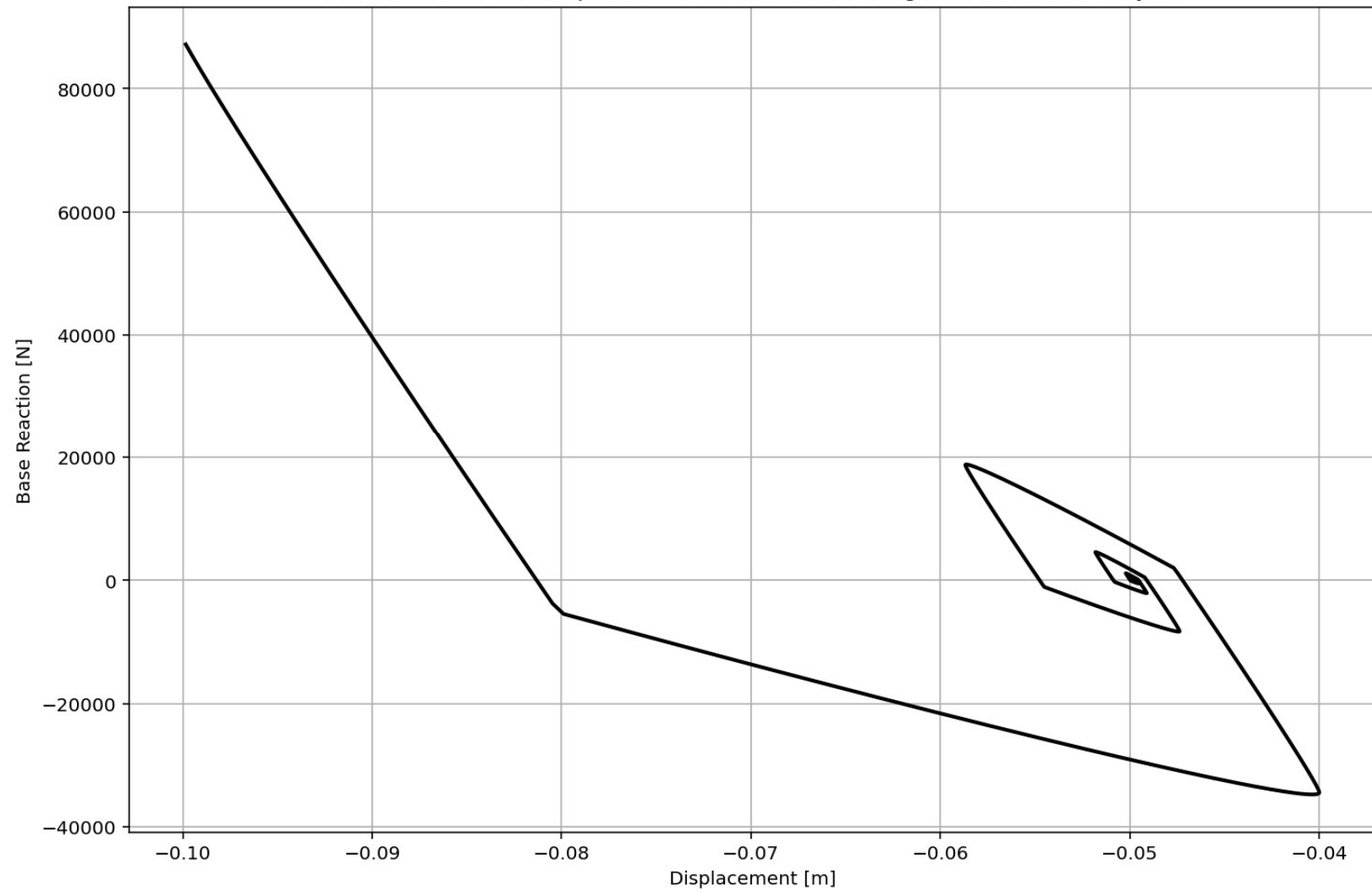


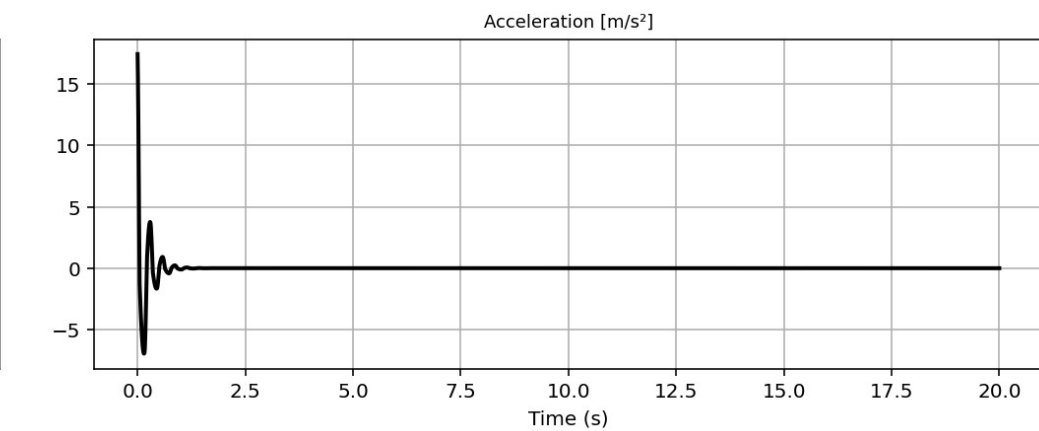
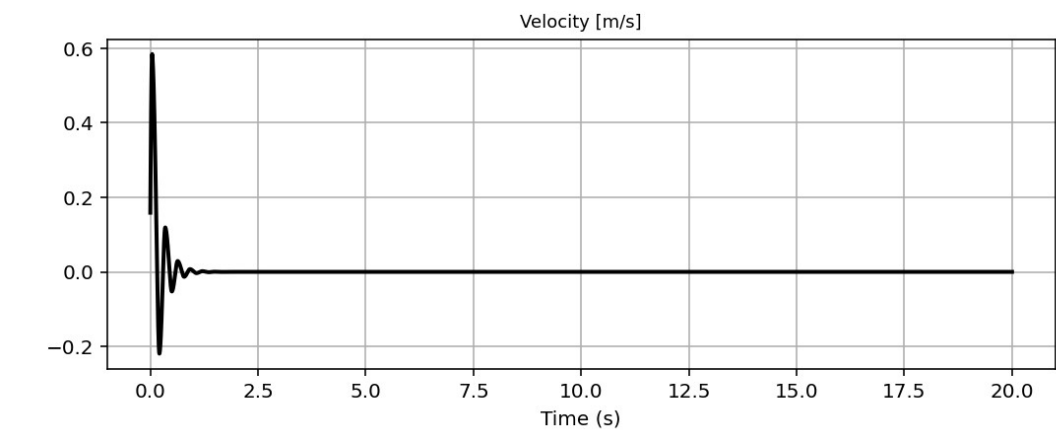
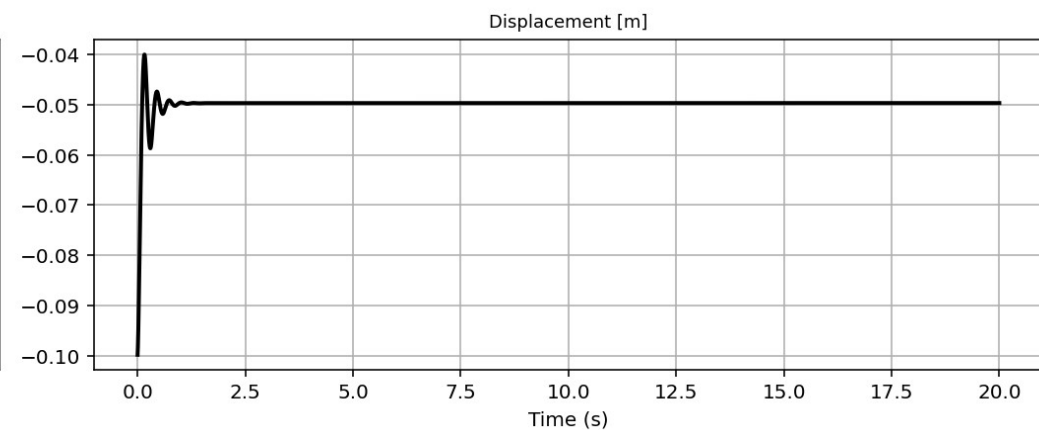
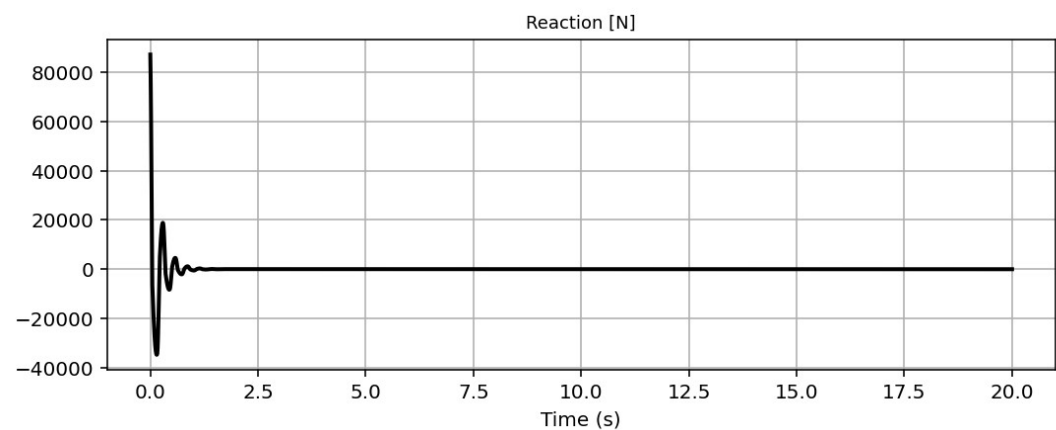


# FREE-VIBRATION ANALYSIS

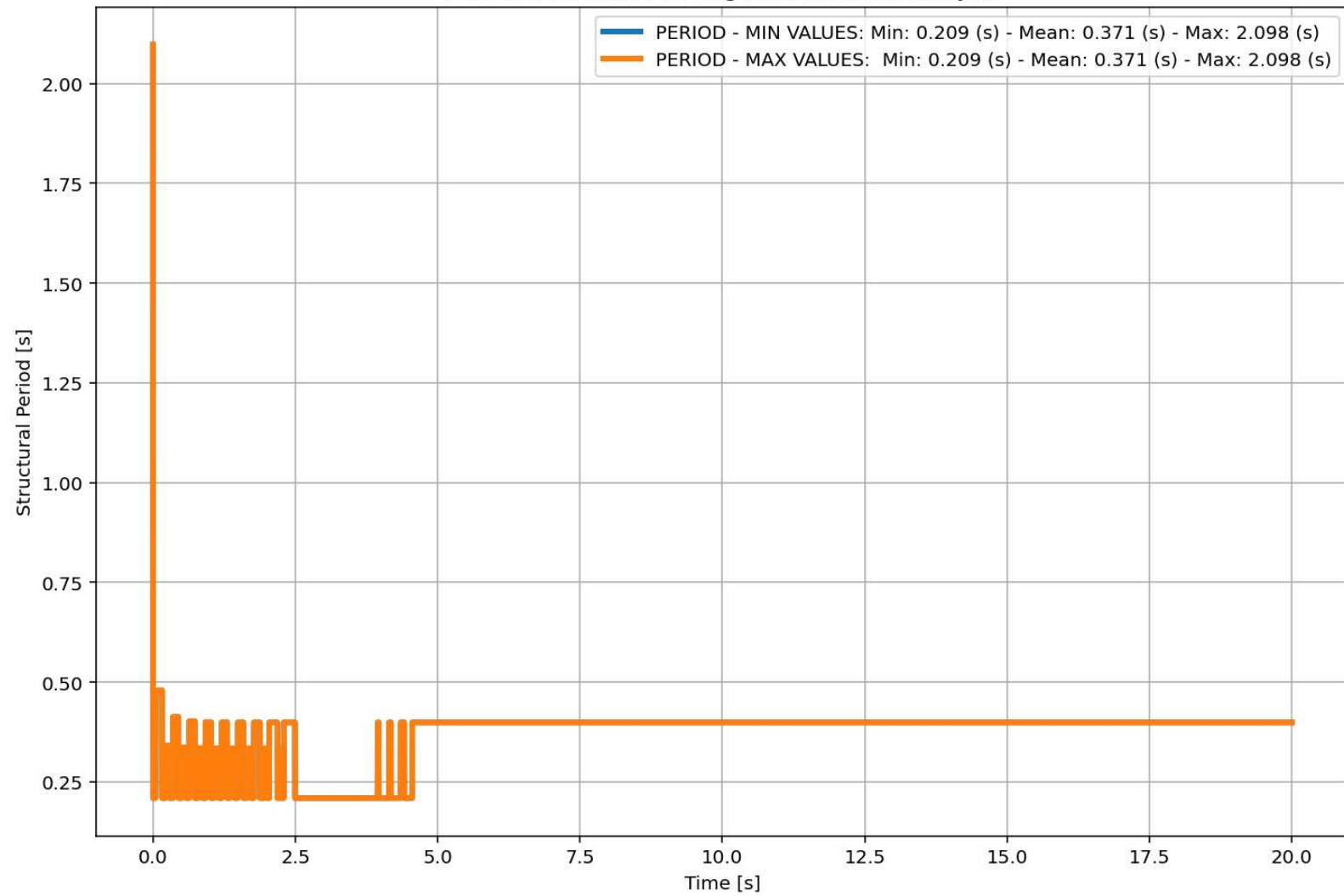


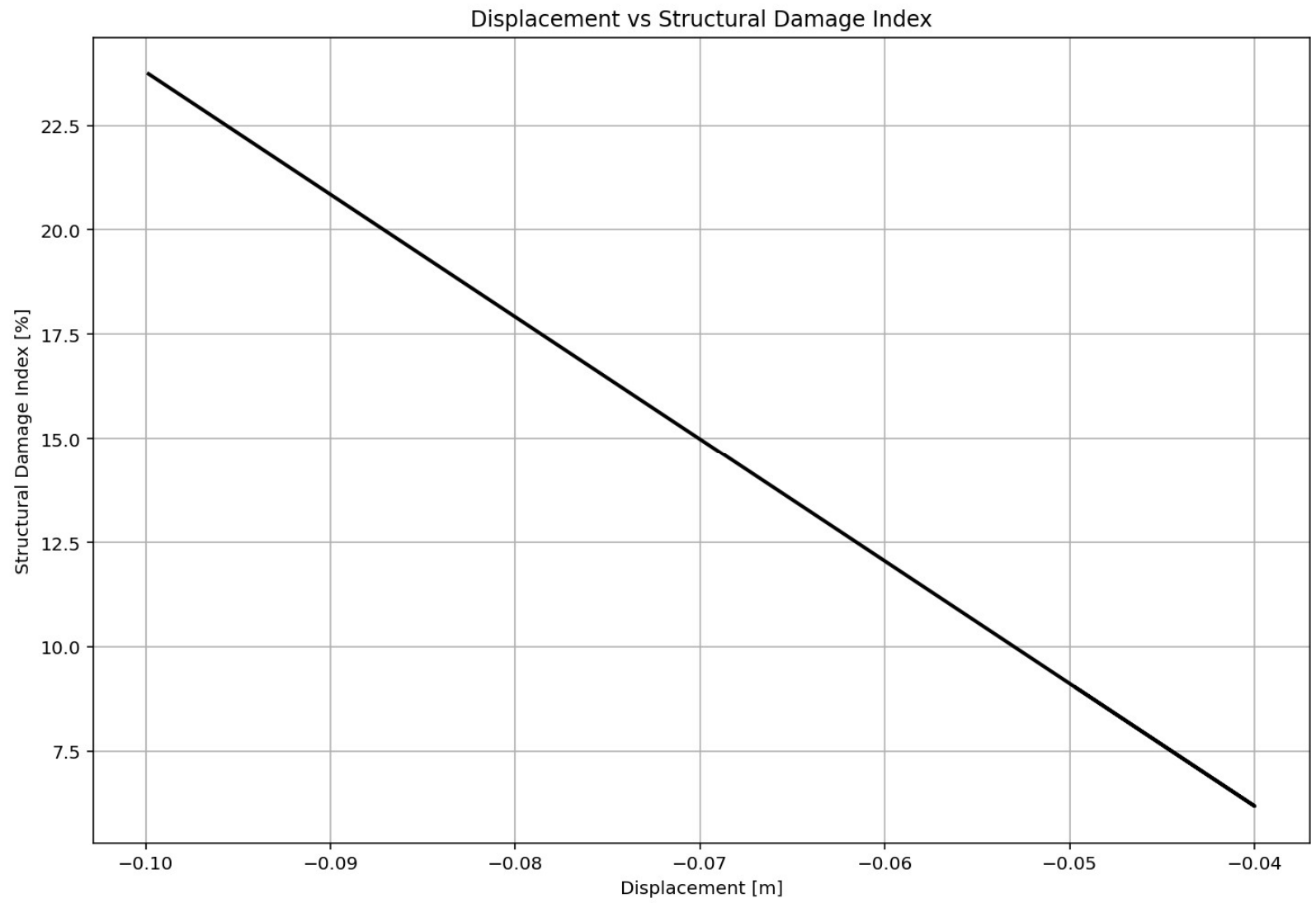
Base Reaction and Dispalcement of Structure During Free-vibration Analysis

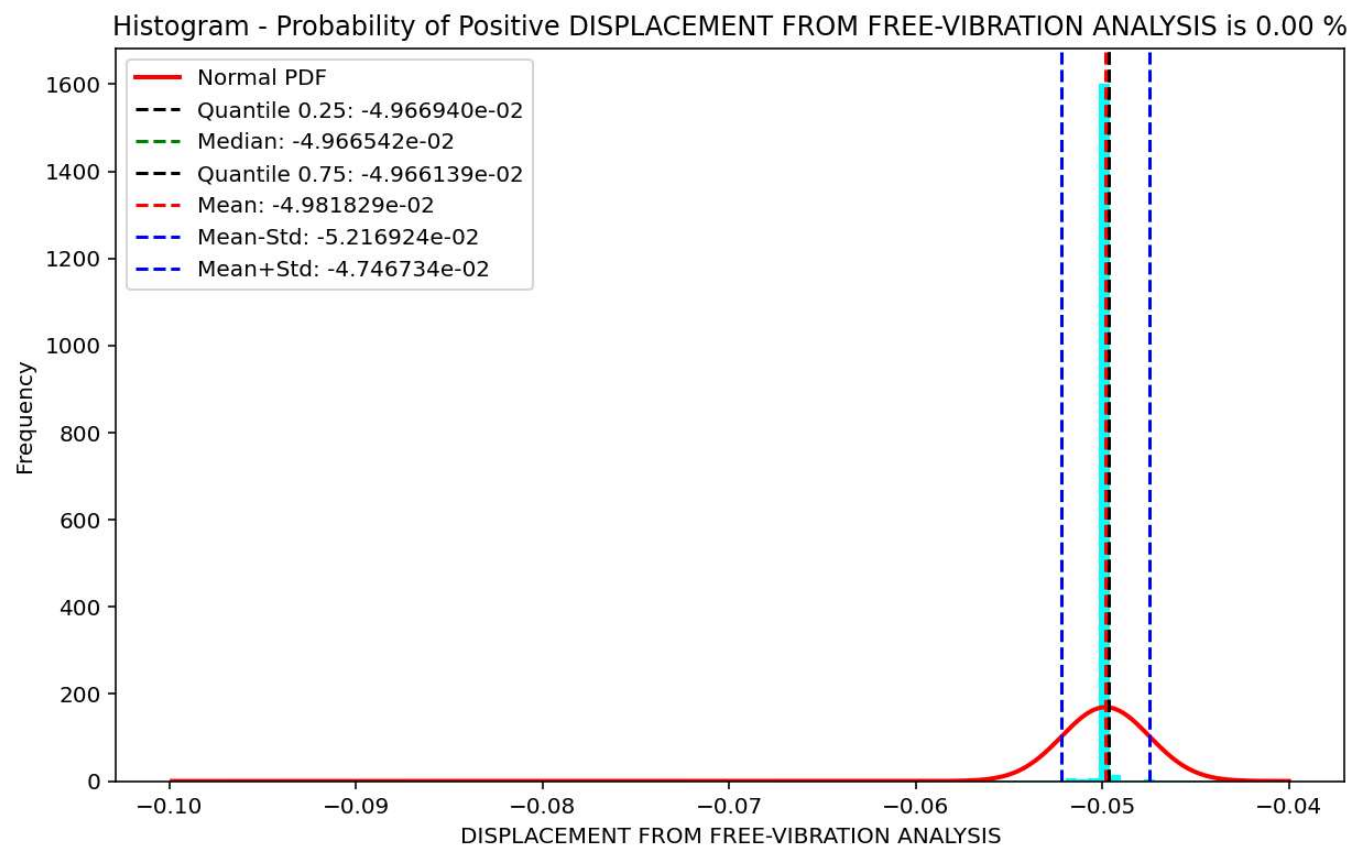




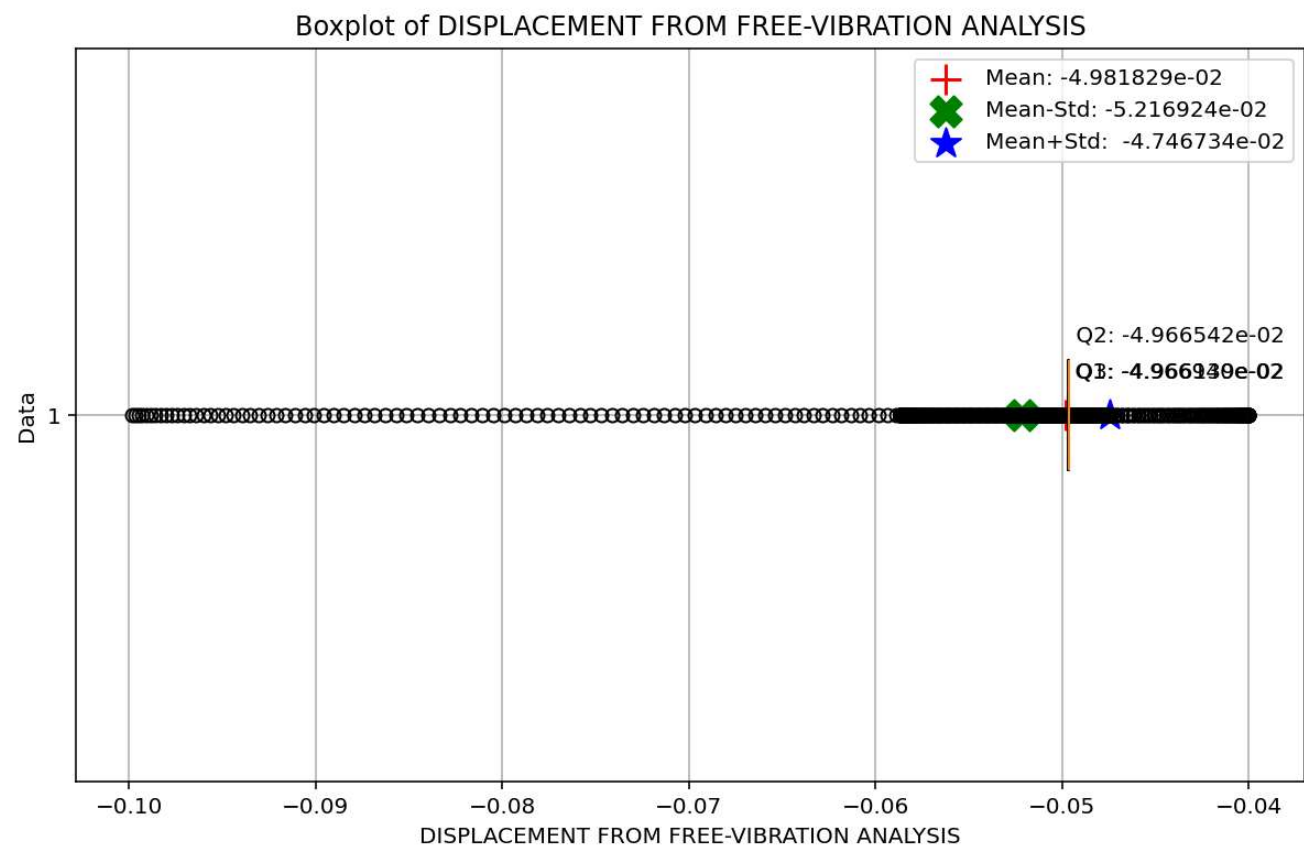
Period of Structure During Free-vibration Analysis

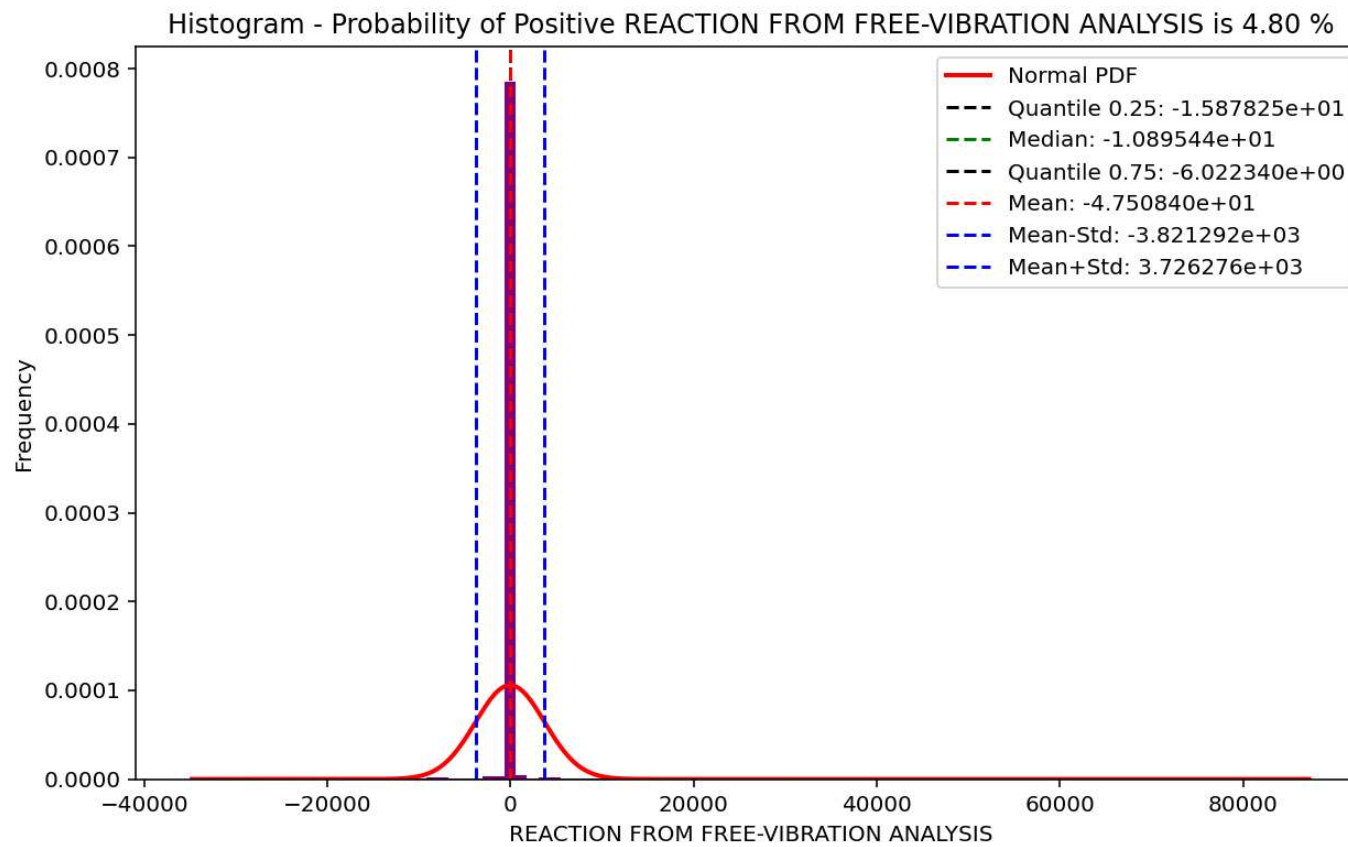


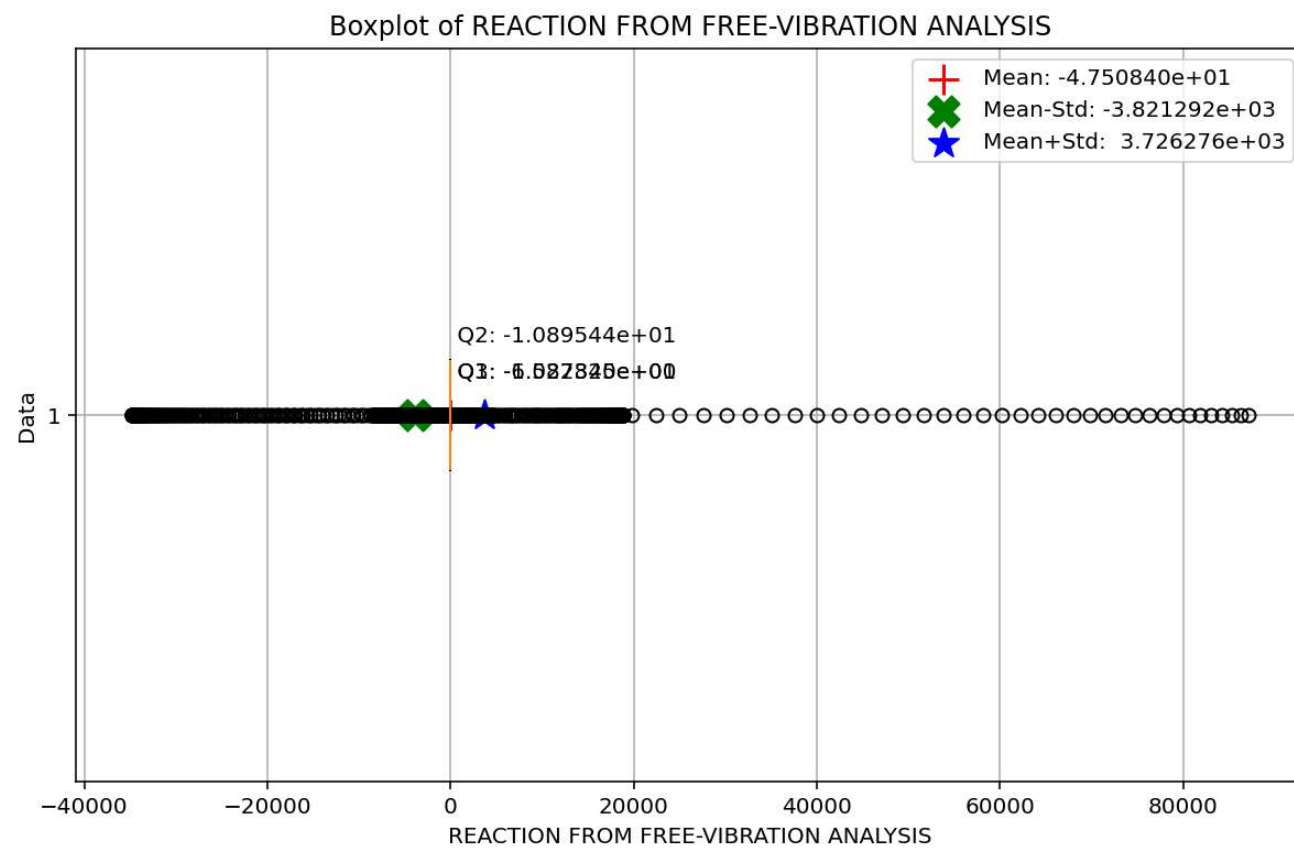




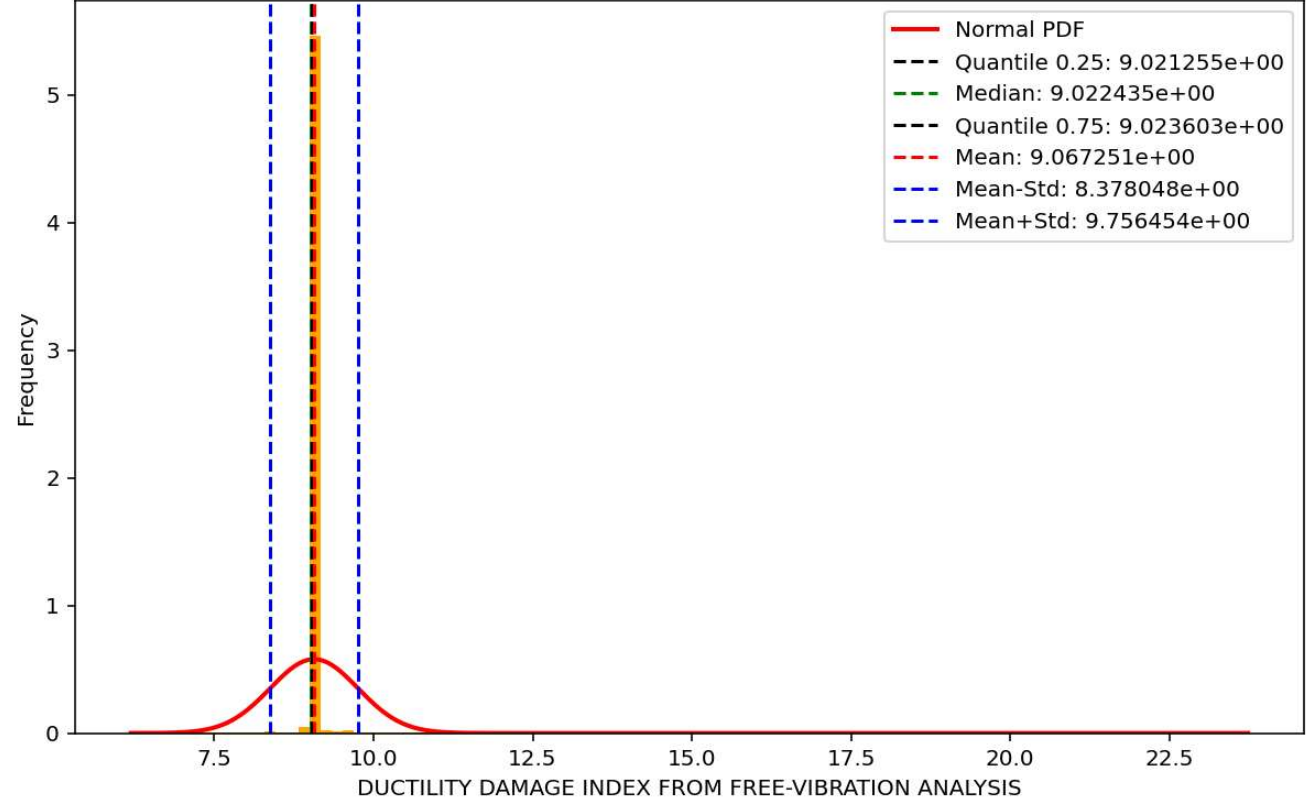


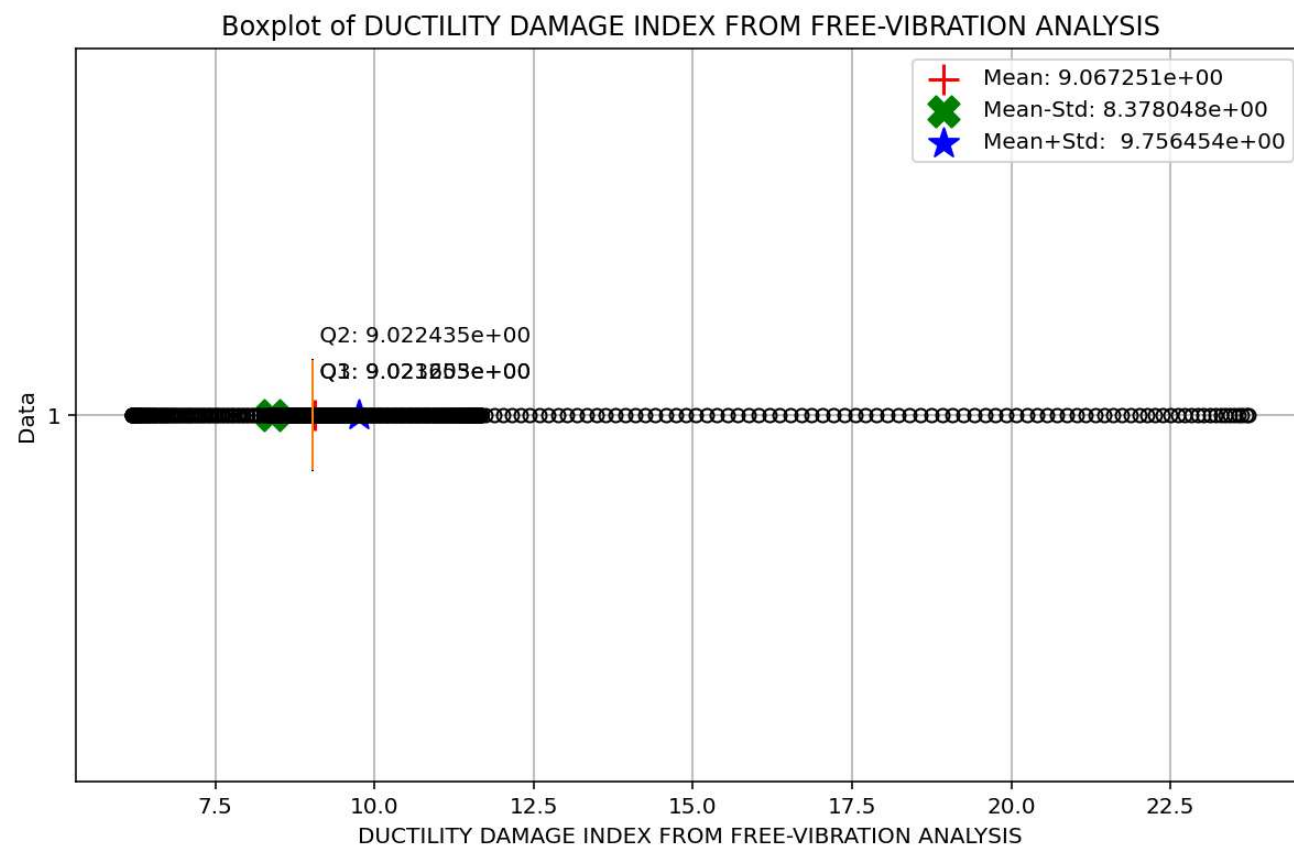




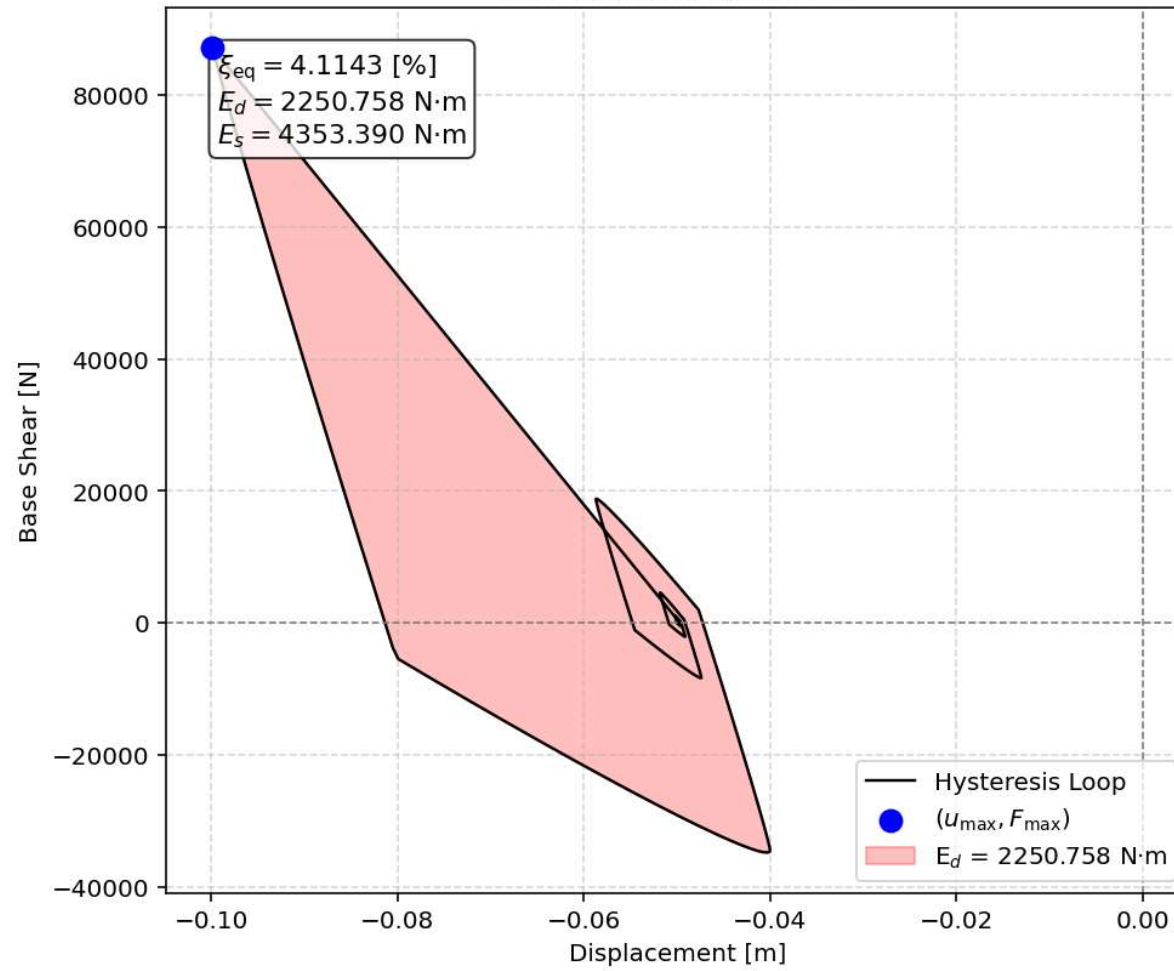


Histogram - Probability of Positive DUCTILITY DAMAGE INDEX FROM FREE-VIBRATION ANALYSIS is 100.00 %





Free-Vibration Hysteresis

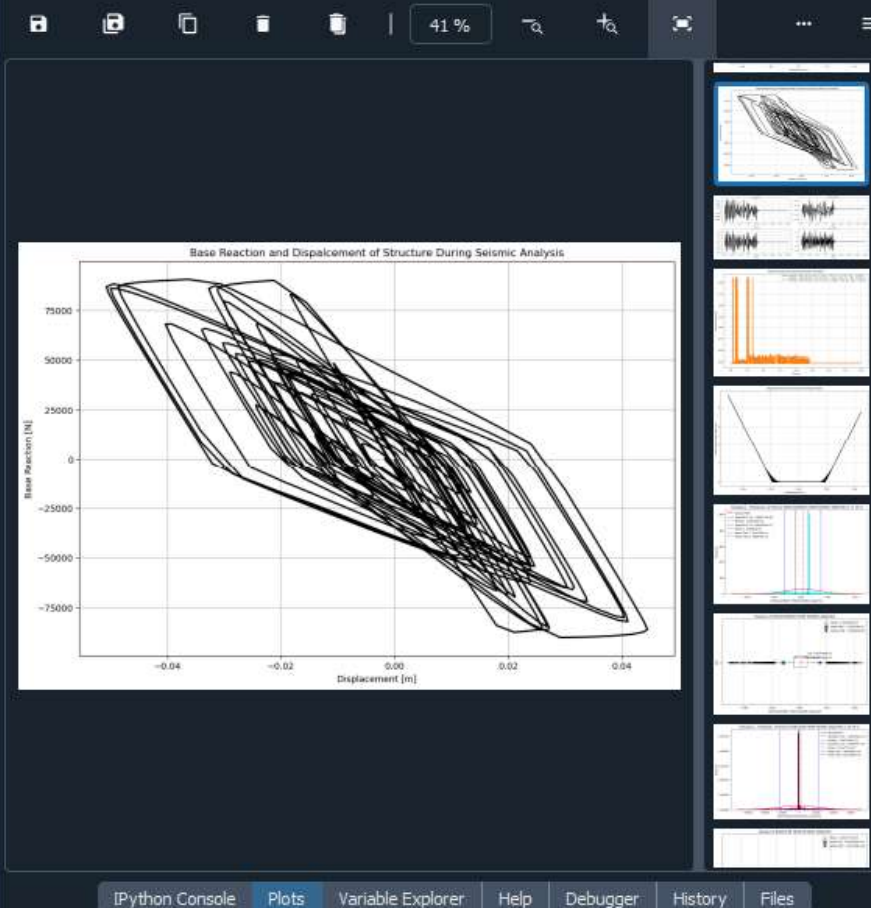


# SEISMIC ANALYSIS

C:\Users\Dell\Desktop\OPENSEES\_FILES\+EIGHT\_ANALY...X\11\_SDOF\_8\_ANA\+11\_SDOF\_8\_ANA\P(t)\_SDOF\_8\_ANA.py

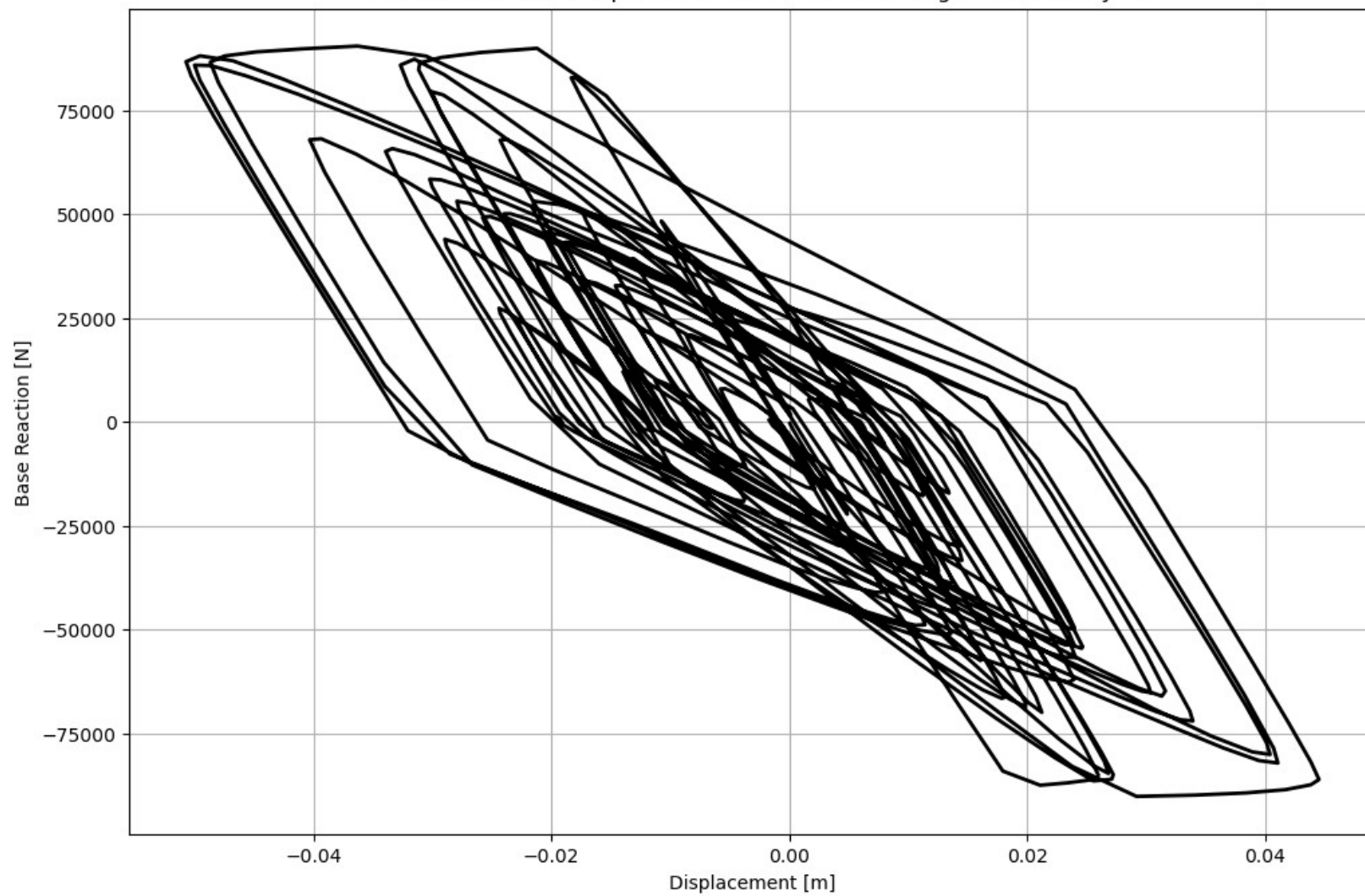
P(t)\_SDOF\_8\_ANA.py X

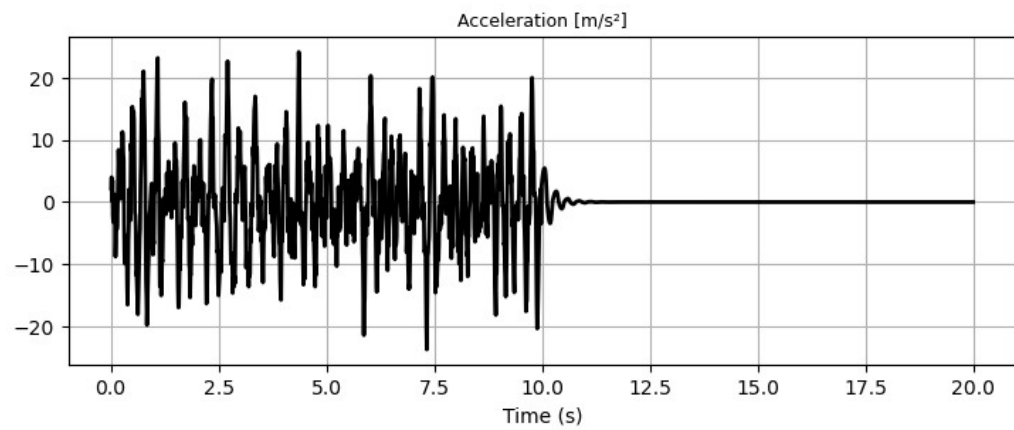
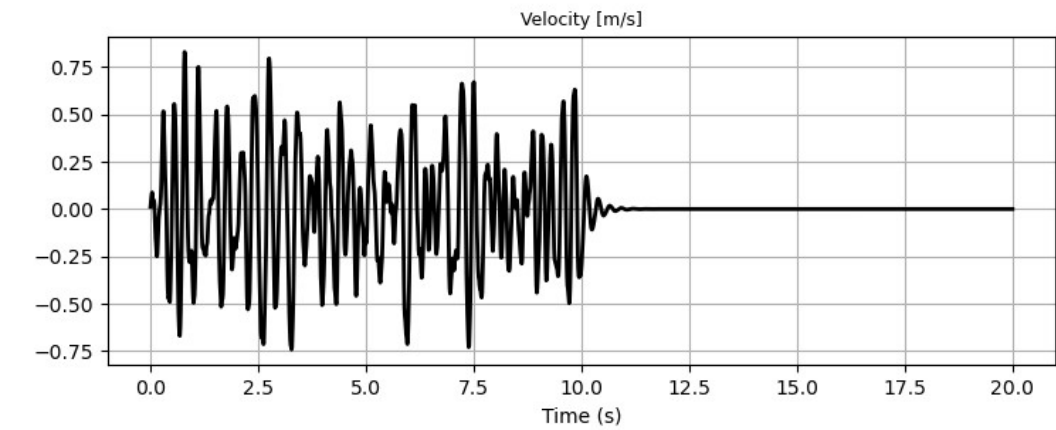
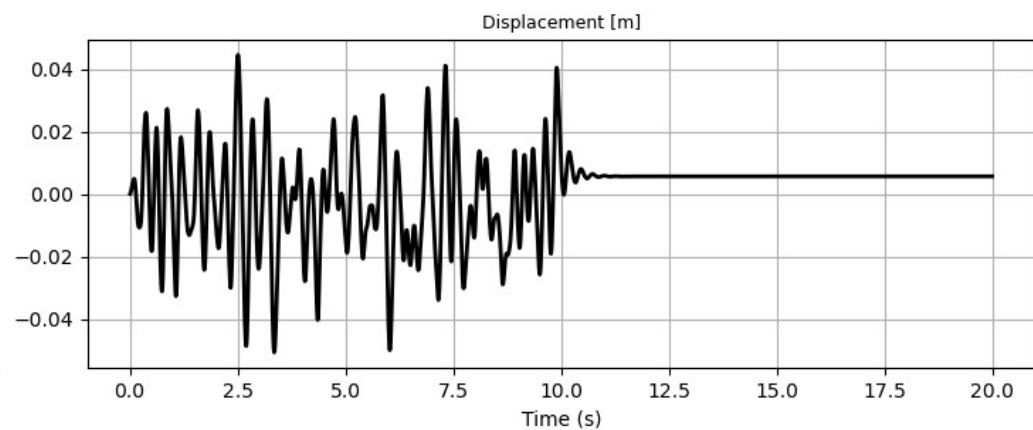
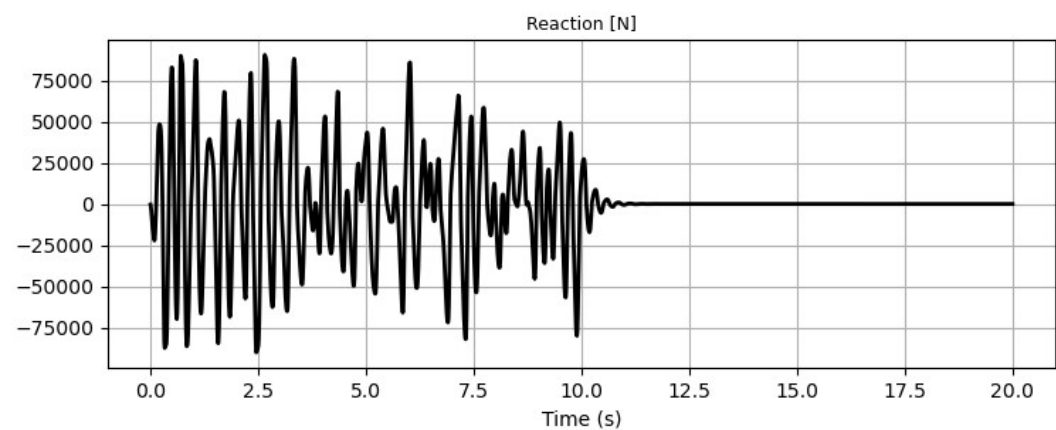
```
1077 # SEISMIC ANALYSIS (DYNAMIC TIME-HISTORY ANALYSIS)
1078 MAT_TYPE = 'INELASTIC' # 'ELASTIC' OR 'INELASTIC'
1079 ANAL_TYPE = 'SEISMIC'
1080
1081 DATA = SDOF(MAT_TYPE, TOTAL_MASS, ANAL_TYPE)
1082
1083 (time_SEI, reaction_SEI, disp_SEI, velo_SEI, acc_SEI, DI_SEI,
1084  stiffness_SEI, PERIOD_SEI, damping_ratio_SEI,
1085  PERIOD_MIN_SEI, PERIOD_MAX_SEI) = DATA
1086
1087
1088 XDATA = disp_SEI
1089 YDATA = reaction_SEI
1090 XLABEL = 'Displacement [m]'
1091 YLABEL = 'Base Reaction [N]'
1092 TITLE = 'Base Reaction and Dispalcement of Structure During Seismic Analysis'
1093 COLOR = 'black'
1094 SEMILOGY = False
1095 PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMILOGY)
1096
1097 PLOT_TIME_HISTORY(time_SEI, reaction_SEI, disp_SEI, velo_SEI, acc_SEI)
1098
1099 # PLOT STRUCTURAL PERIOD DURING THE ANALYSIS
1100 plt.figure(0, figsize=(12, 8))
1101 plt.plot(time_SEI, PERIOD_MIN_SEI, linewidth=3)
1102 plt.plot(time_SEI, PERIOD_MAX_SEI, linewidth=3)
1103 plt.title('Period of Structure During Seismic Analysis')
1104 plt.ylabel('Structural Period [s]')
1105 plt.xlabel('Time [s]')
1106 #plt.semilogy()
1107 plt.grid()
1108 plt.legend([f'PERIOD - MIN VALUES: Min: {np.min(PERIOD_MIN_SEI):.3f} (s) - Mean: {np.mean(PERIOD_MIN_SEI):.3f} (s)',
1109            f'PERIOD - MAX VALUES: Min: {np.min(PERIOD_MAX_SEI):.3f} (s) - Mean: {np.mean(PERIOD_MAX_SEI):.3f} (s)'])
1110
```

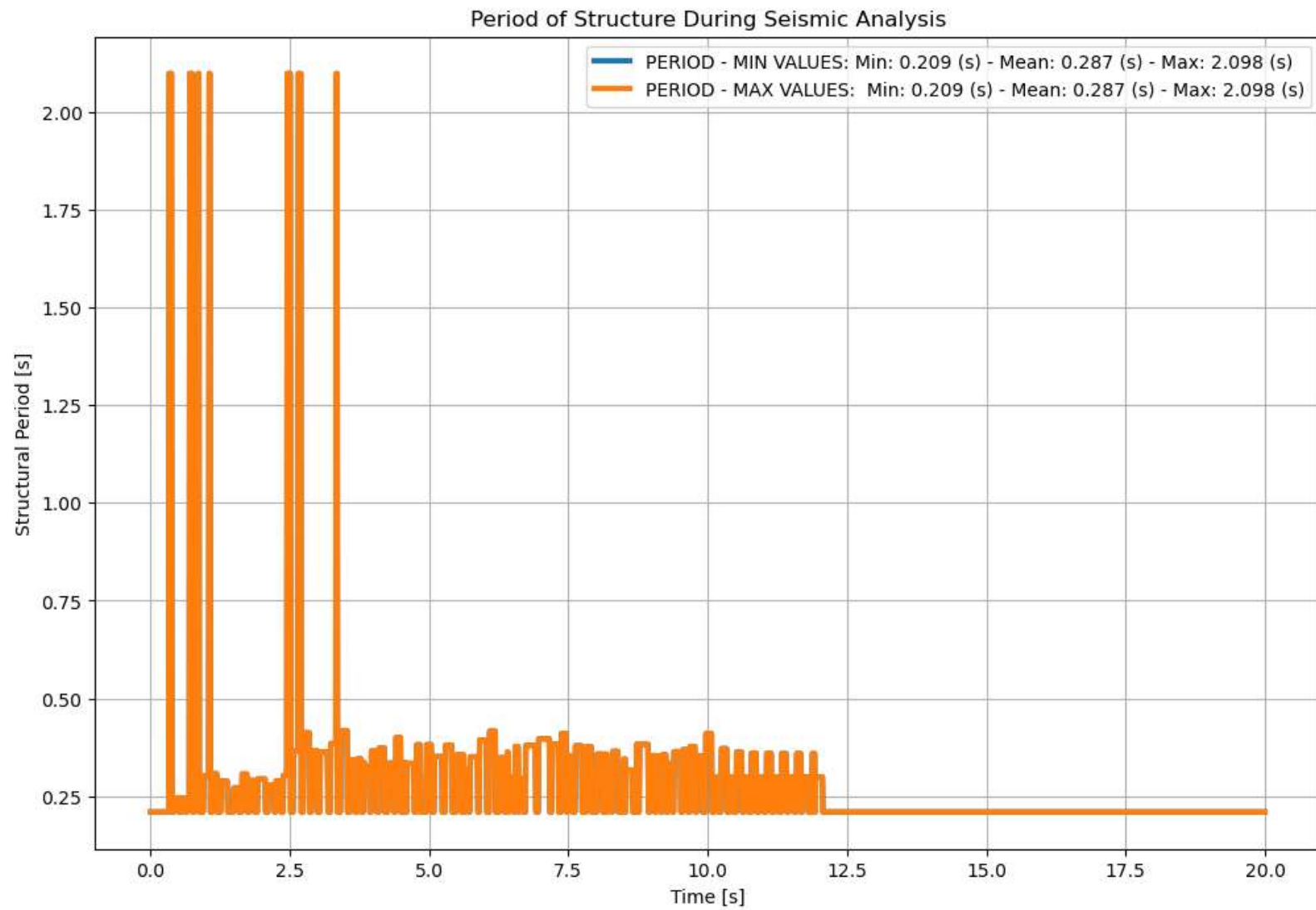


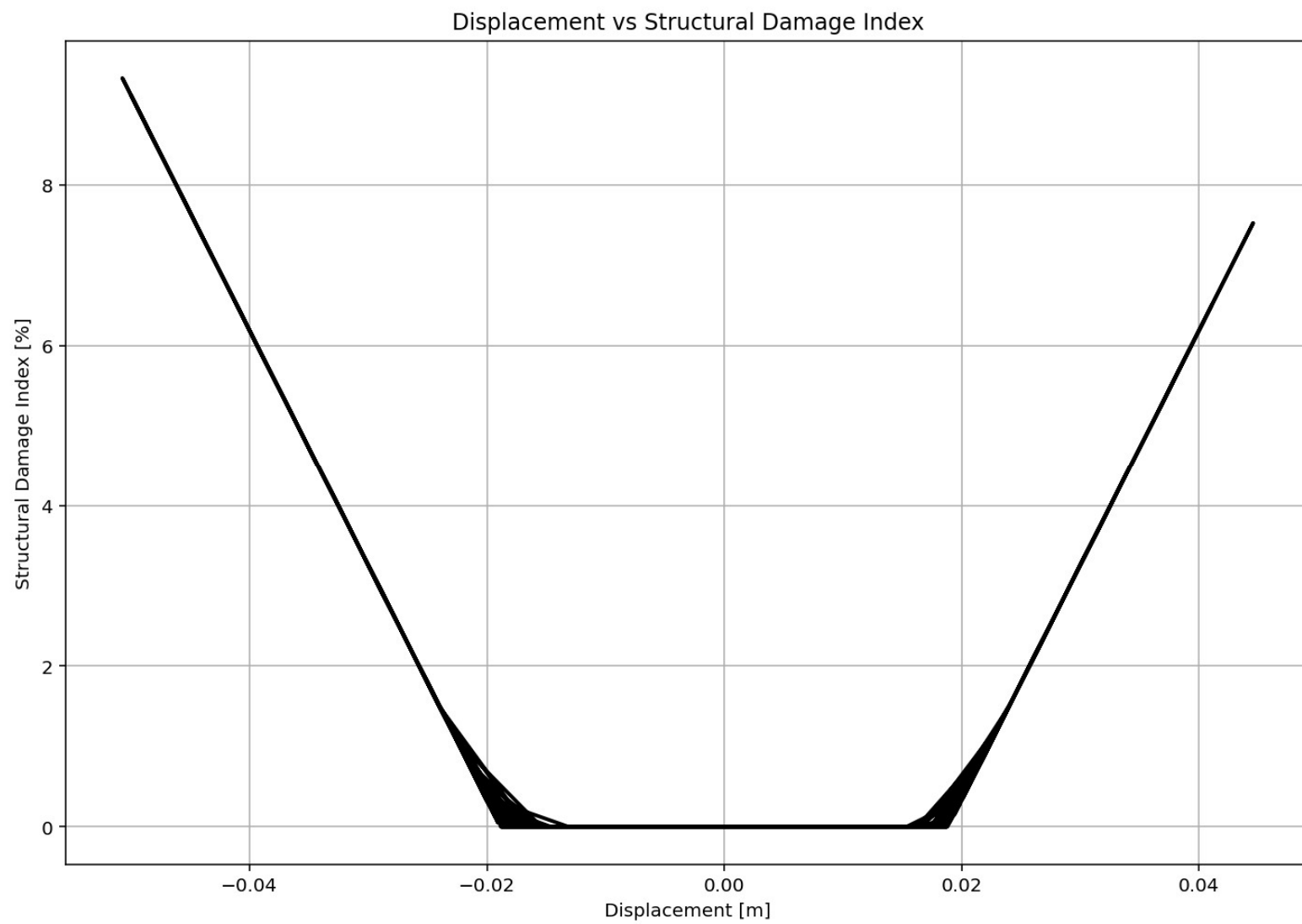


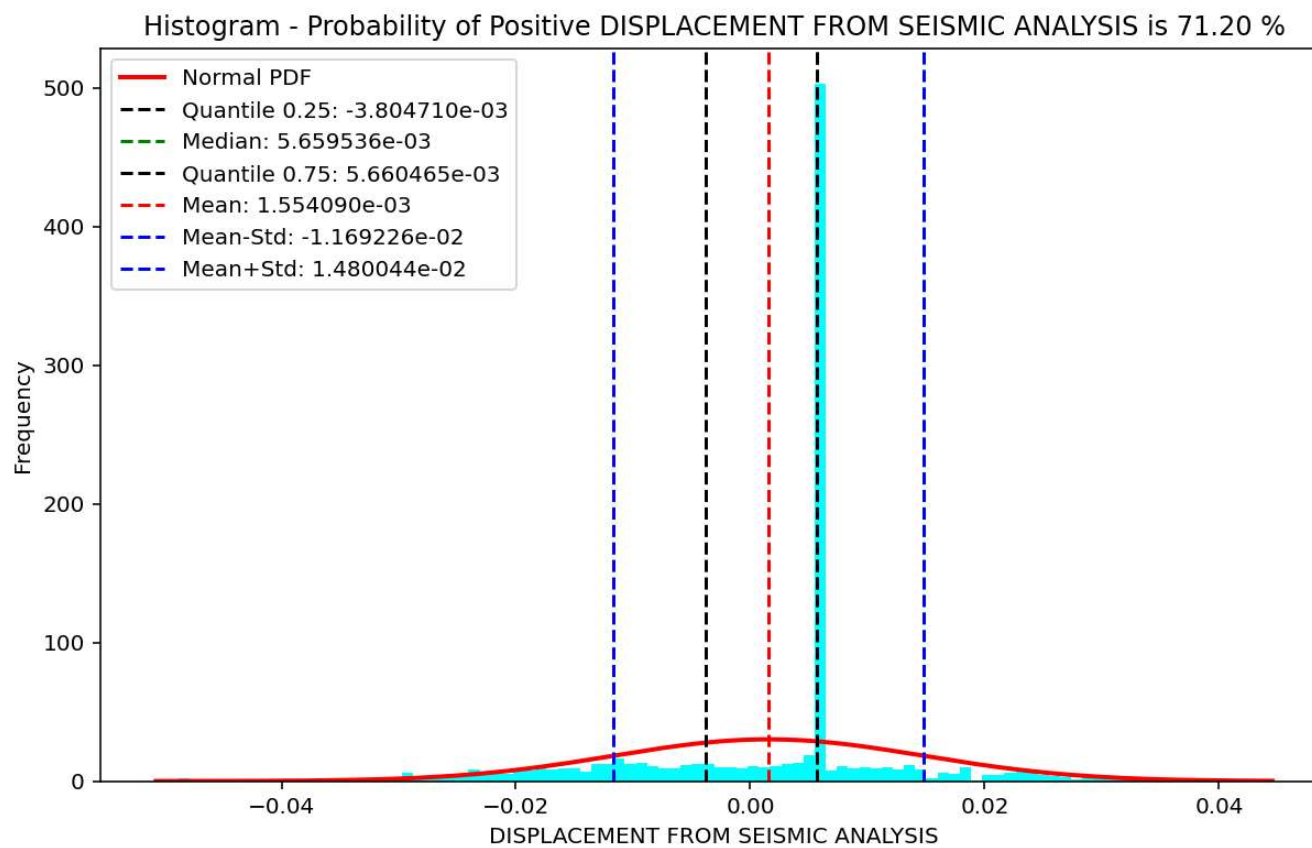
Base Reaction and Dispalcement of Structure During Seismic Analysis

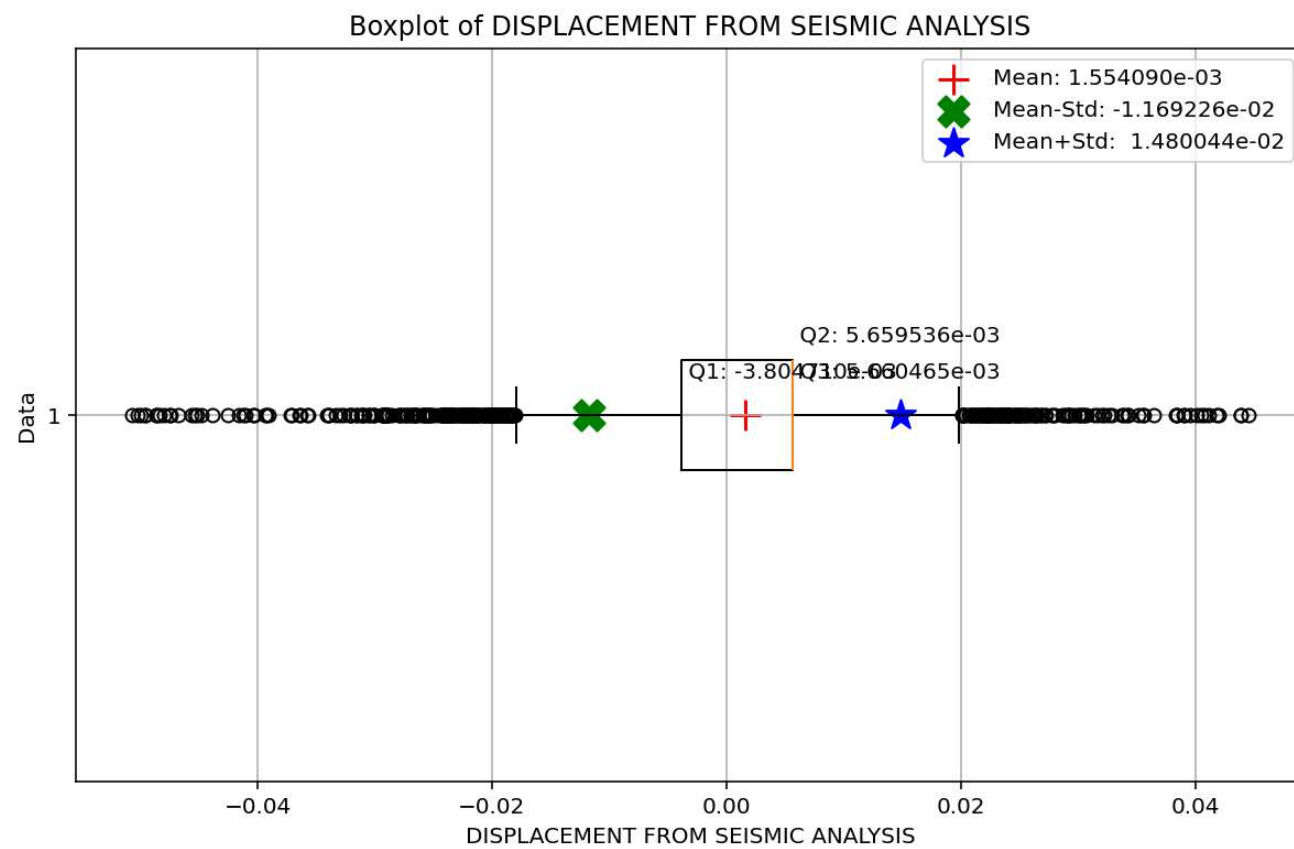


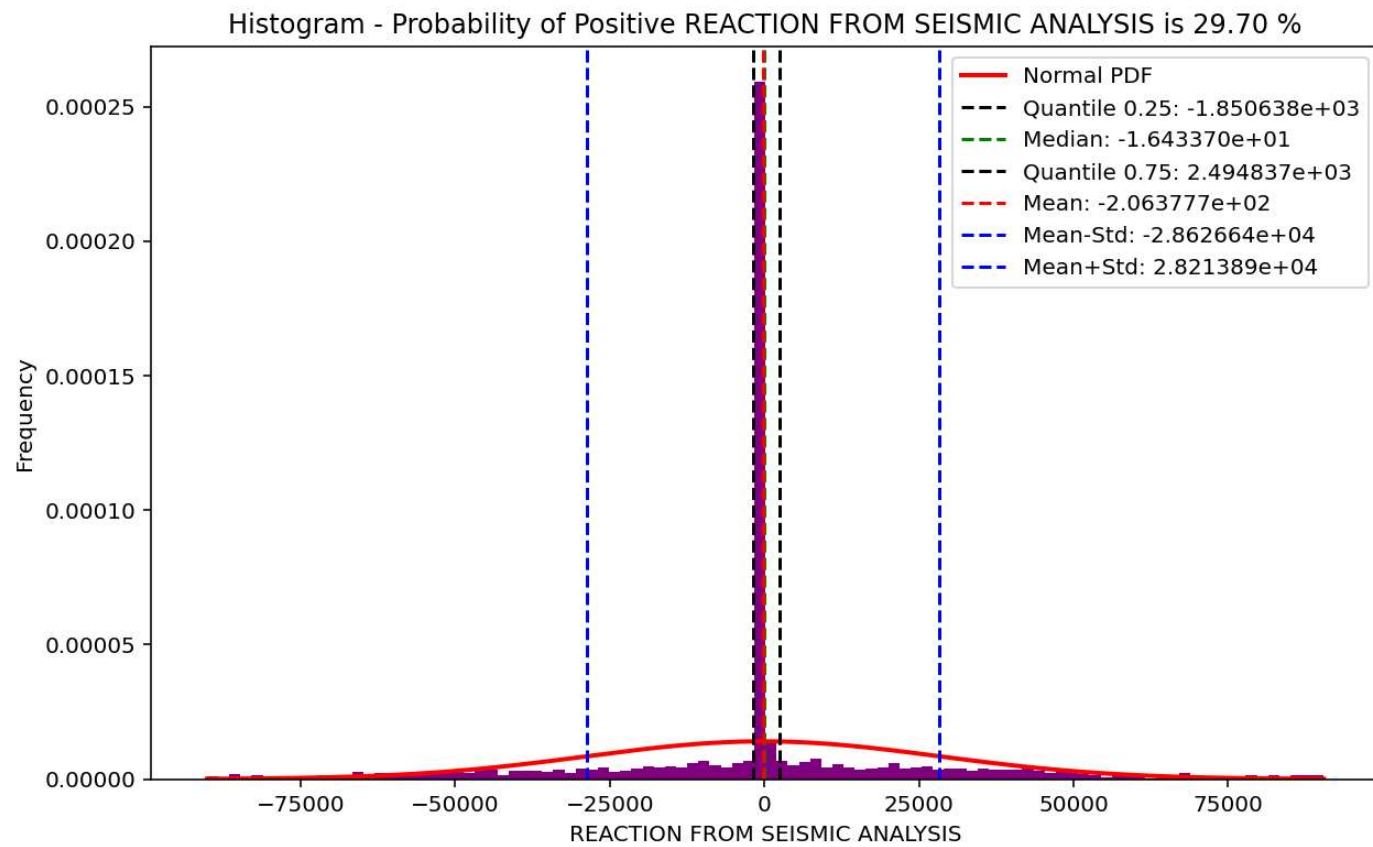


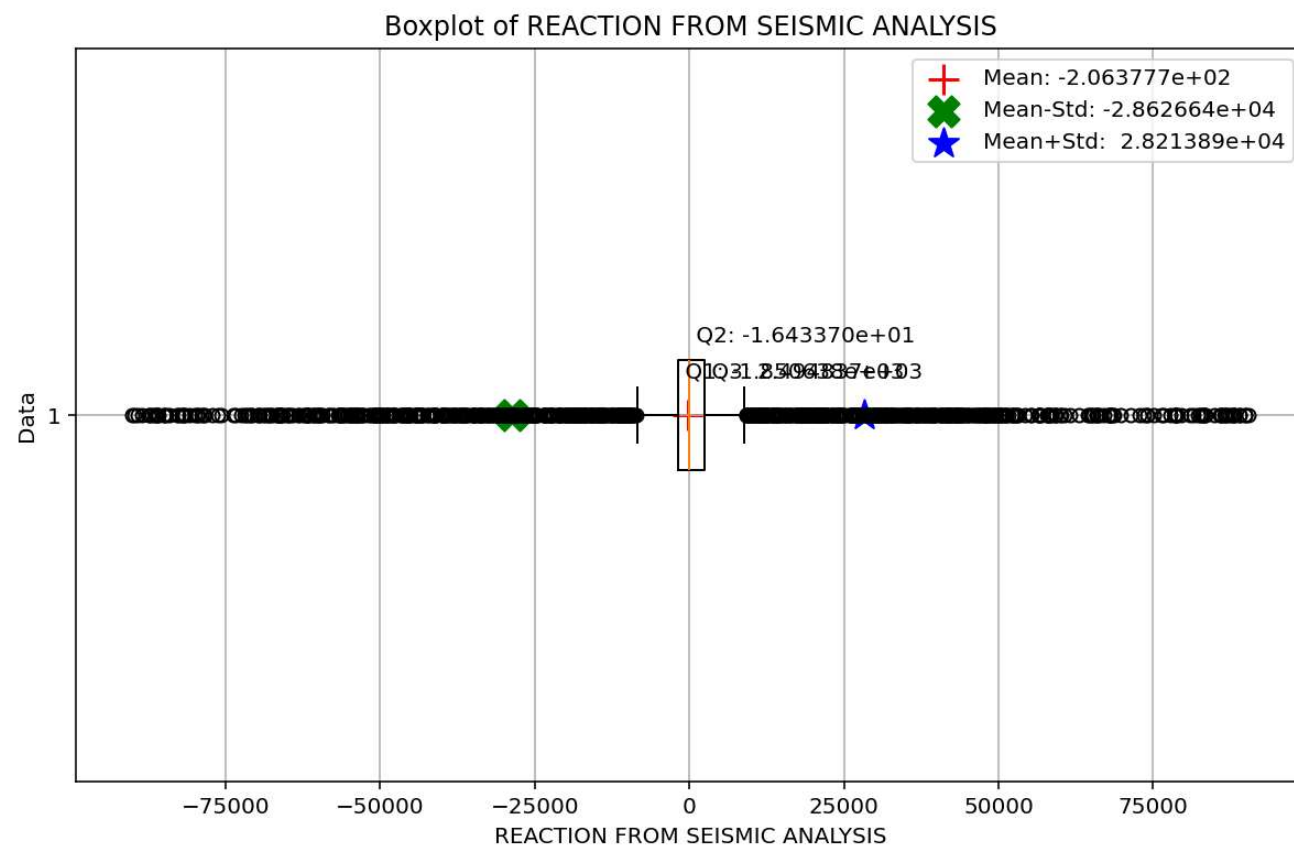






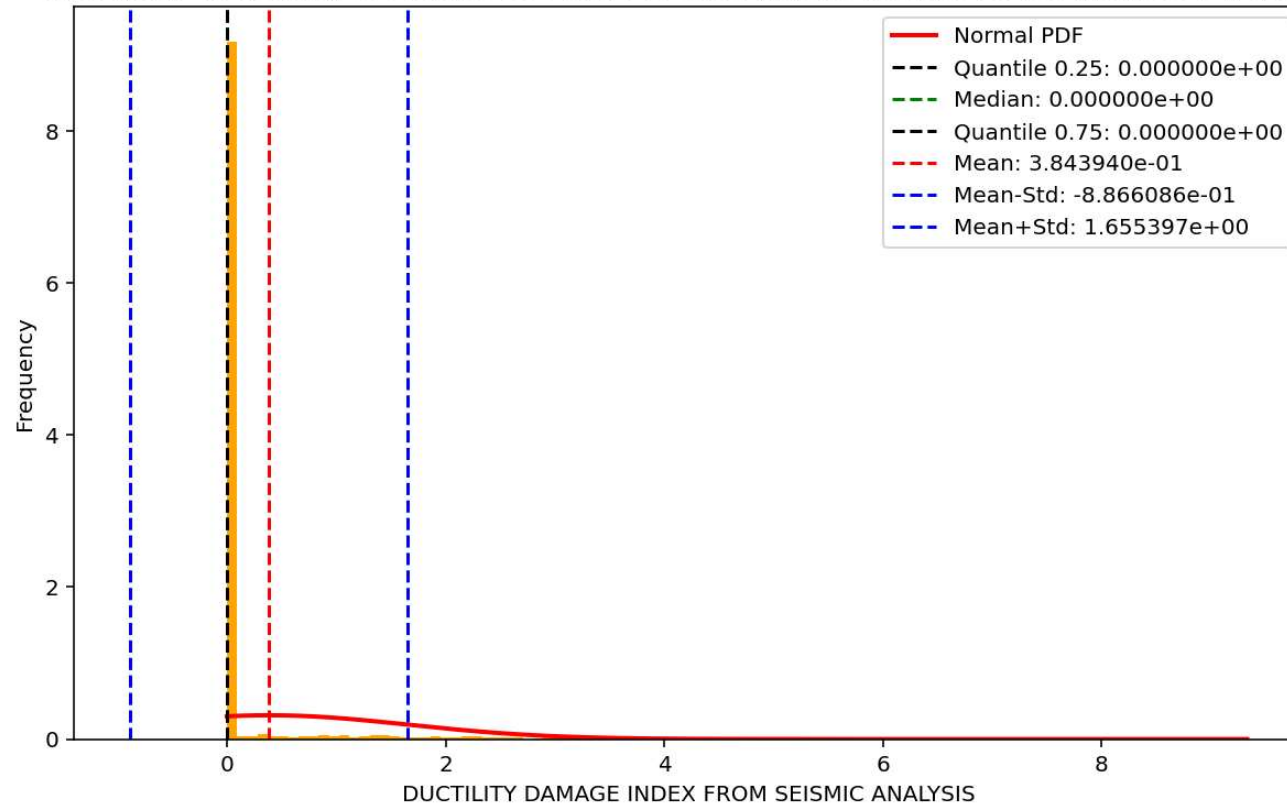


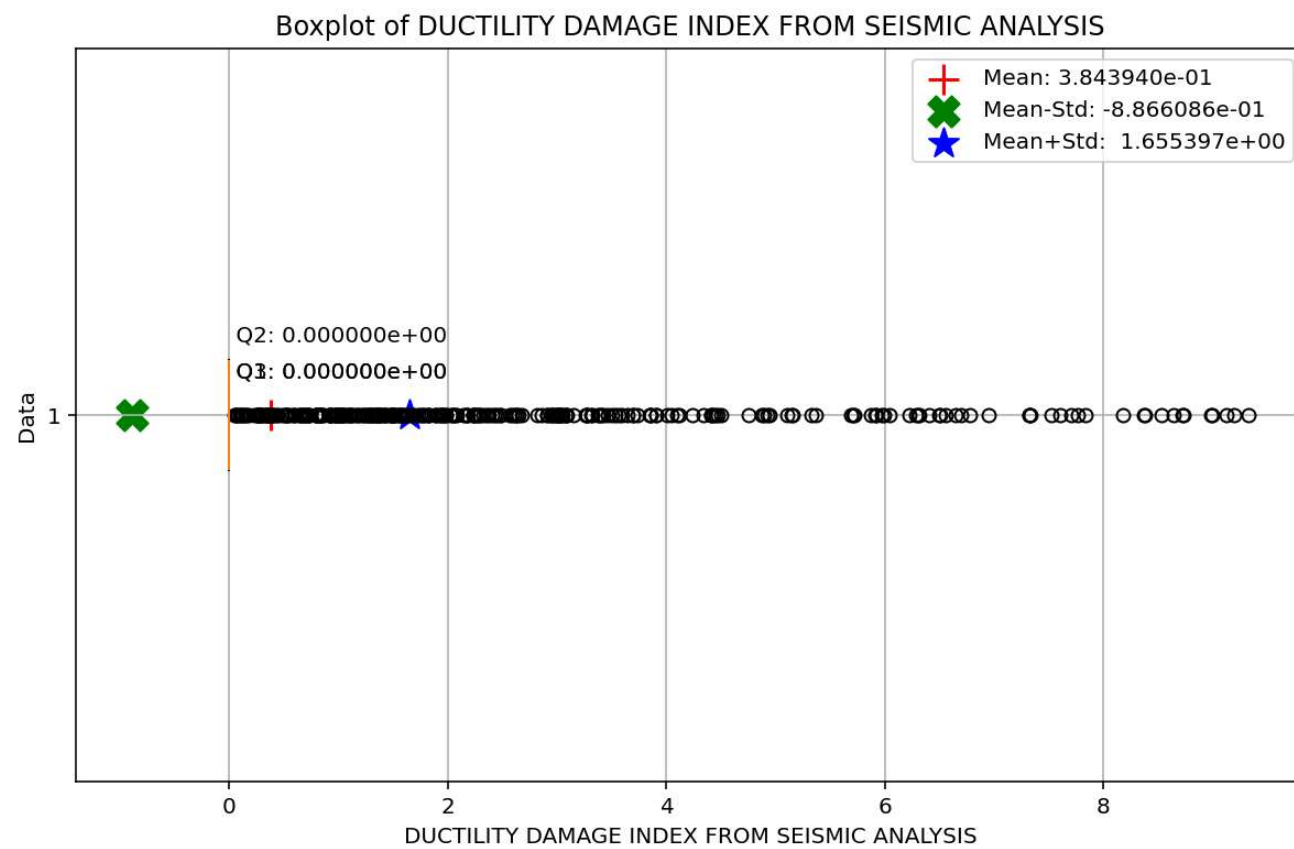


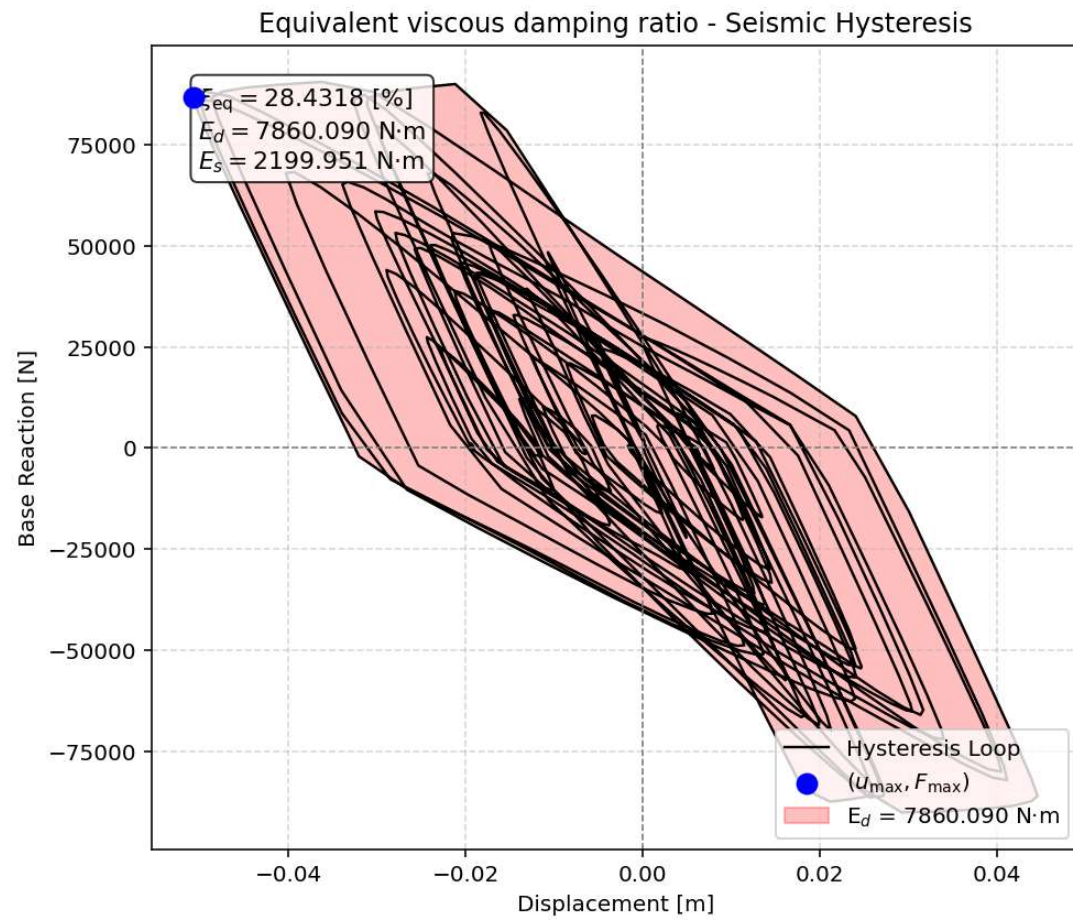




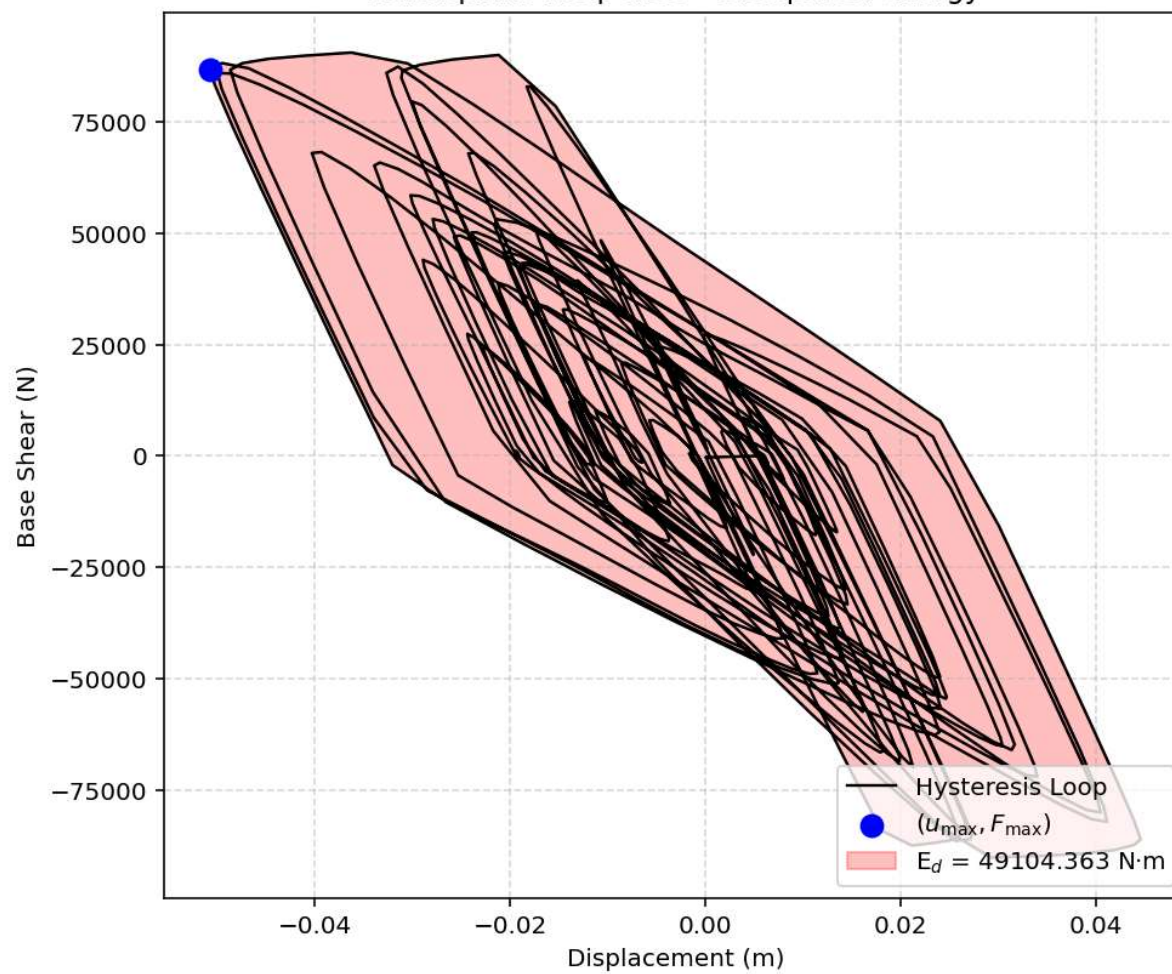
Histogram - Probability of Positive DUCTILITY DAMAGE INDEX FROM SEISMIC ANALYSIS is 14.60 %

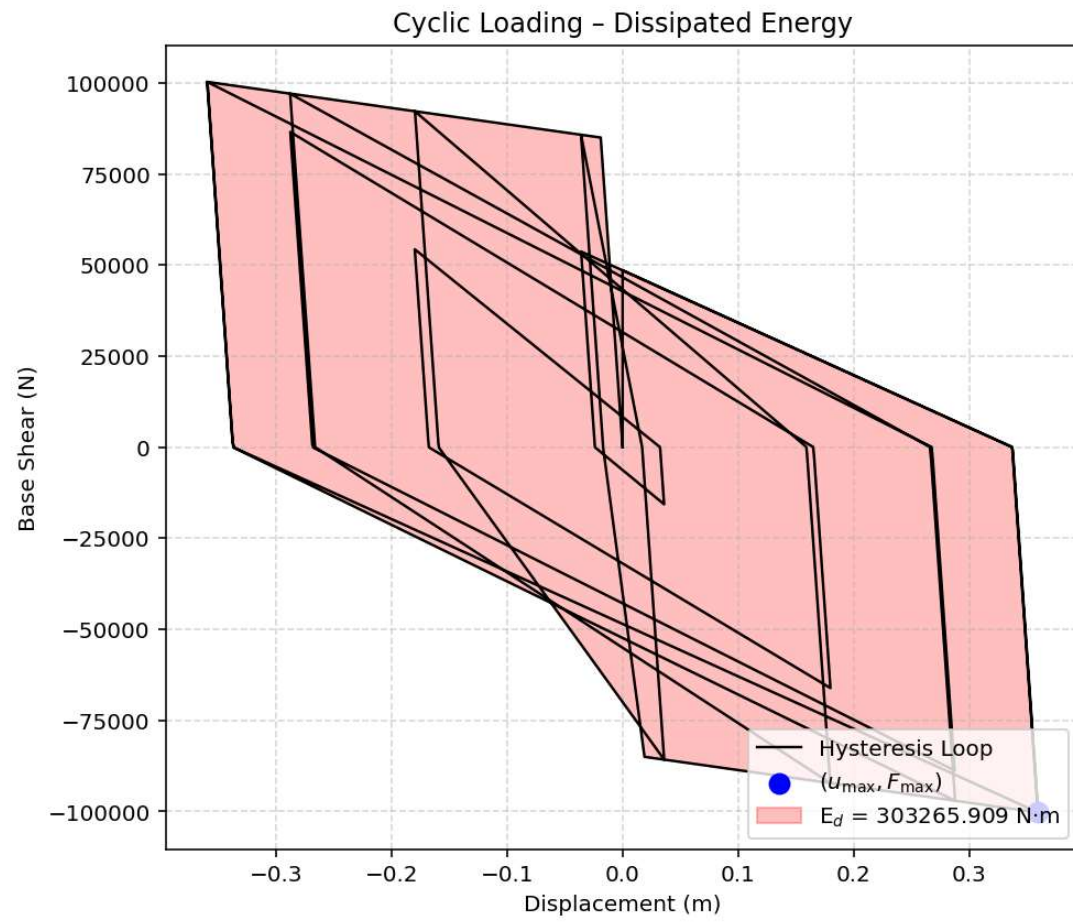






Earthquake Response - Dissipated Energy

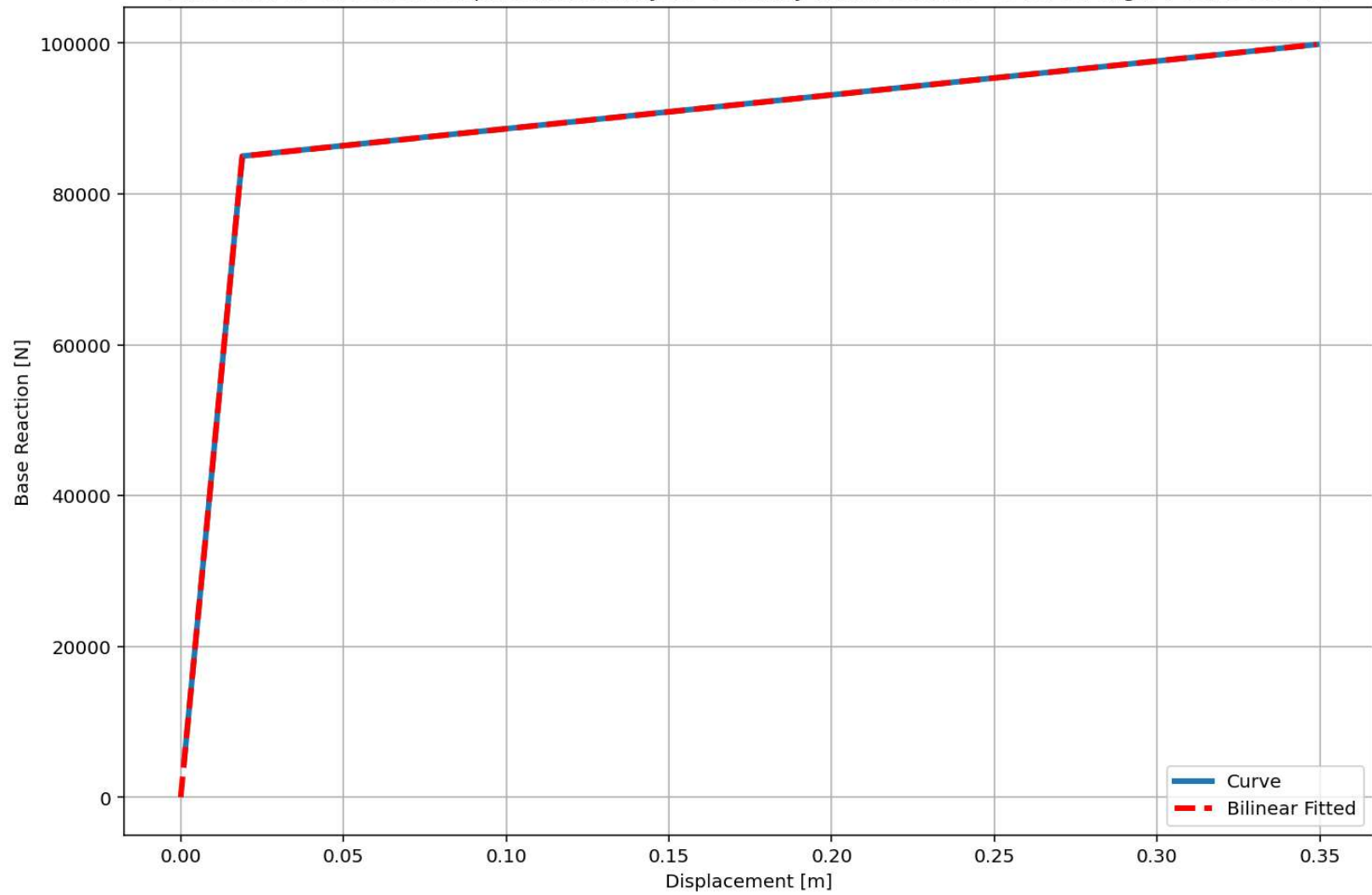




**Dissipated Energy from Earthquake= 49104.36 N·m**  
**Dissipated Energy from Cyclic Displacement= 303265.91 N·m**

**DISSIPATED ENERGY CAPACITY INDEX: 16.192 [%]**  
**ZONE 2: MINOR DAMAGE**

Last Data of BaseShear-Displacement Analysis - Ductility Ratio: 18.4798 - Over Strength Factor: 1.1744



**Equivalent viscous damping ratio = 28.4318**

**+=====+**

**= Analysis curve fitted =**

**Disp Base Shear**

**-----**

**[[0.00000000e+00 0.00000000e+00]**

**[1.88856382e-02 8.49853721e+04]**

**[3.60000000e-01 1.00300000e+05]]**

**+=====+**

**+-----+**

**Structure Elastic Stiffness : 4500000.00**

**Structure Plastic Stiffness : 278611.11**

**Structure Tangent Stiffness : 44895.88**

**Structure Ductility Ratio : 19.06**

**Structure Over Strength Factor: 1.18**

**+-----+**

**Over Strength Coefficient ( $\Omega_0$ ): 1.1802**

**Displacement Ductility Ratio ( $\mu$ ): 19.0621**

**Ductility Coefficient ( $R_\mu$ ): 19.0621**

**Structural Behavior Coefficient (R): 22.4972**

**Structural Ductility Damage Index in X Direction: 9.3315 (%)**



Seismic Fragility Curves by Damage State from DYN. TIME-HISTORY ANA. DUCTILITY DAMAGE INDEX [%]

